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THE PENNY MAGAZINE OF THE SOCIETY FOR THE DIFFUSION OF USEFUL KNOWLEDGE.



New Series.

1843.

CHARLES KNIGHT & CO.,
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THE PENNY MAGAZINE

OF THE

Society for the Diffusion of Useful Knowledge.



[Sir Roger de Coverley and the 'Spectator.']

SIR ROGER DE COVERLEY.—No. I.

It is pleasant to reflect upon the imperishable quality of many of those things, apparently trifling, which have the power of contributing to innocent enjoyment. The sports of childhood are essentially ancient. The top and the hoop have outlived many generations. There is a famous picture by Lionardo da Vinci, in which a boy is playing with the pretty toy in which a number of flat boards are fastened by tapes—at once dissevered and united; and the toy is still sold for a halfpenny at the corner of every street. To ascend in the scale of enjoyment, the melody which was delightful in the days of Queen Elizabeth is forgotten, perhaps, for two hundred years, and it suddenly springs into popularity in the days of Queen Victoria. For a quarter of a century country-dances were out of fashion. They are reviving; and with them comes back one of the oldest and most beautiful, with its courteous advances, from the extremities of a long line, of the lady and the gentleman,—their turnings in the centre,—their returnings,—the chain figure in which the lady winds through a line of gentlemen, and the gentleman through a line of ladies—and lastly, the arched hands under which every couple passes. This is Roger de Coverley, or Roger of Cowley. Cowley is a pretty village about two miles from Oxford; and here some one lived in the days of the Tudors who was famous enough to have his name linked with the pretty dance-tune that has once again become fashionable. But he had a higher honour. The popularity of the dance in the days of Queen Anne gave a name to the most famous character in 'The Spectator;' and ever afterwards the dance itself gathered an accession of dignity even in its name; and plain Roger of Cowley became *Sir* Roger *de* Coverley.

The revival of the dance is propitious to our attempt to revive, for the general reader, those delightful papers of Addison and Steele which are devoted to the fictitious character of Sir Roger. Few people now read 'The Spectator' as a whole. Some of the more celebrated essays, such as 'The Vision of Mirza,' find their place in books of extract. The delicate humour of the delineation of Sir Roger de Coverley is always referred to as the highest effort of Addison's peculiar genius; but not many will take the pains to select

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these sixteen or seventeen papers from the six hundred and thirty which form the entire work. These papers have a completeness about them which show how thoroughly they were written upon a settled plan. Steele appears to have first conceived the character in the second number of 'The Spectator;' but Addison very soon took it out of his friend's hands, who was scarcely able to carry on the portraiture with that refinement which belonged to Addison's conception of the character. Addison, it is said, killed Sir Roger in the fear that another hand would spoil him.

As a representation of manners a century and a half ago, the picture of Sir Roger de Coverley has a remarkable value. The good knight is thoroughly English; and in him we see a beautiful specimen of the old-fashioned gentleman, with a high soul of honour, real benevolence, acute sense, mixed up with the eccentricities which belong to a nation of humourists. The readers of 'The Spectator' are fast diminishing. No one now gives "his days and nights to the volumes of Addison;" but his gentle graceful humour has never been excelled, and nowhere is it more conspicuous than in the papers of which Sir Roger de Coverley is the hero.

The plan of 'The Spectator' is founded upon the fiction of a club that assembles every Tuesday and Thursday to carry on the publication. Sir Roger does not appear highly qualified for a literary colleague—a *collaborateur*, as the French style it,—but he nevertheless is the foremost in 'The Spectator's' "account of those gentlemen who are concerned with me in the work."

"The first of our society is a gentleman of Worcestershire, of an ancient descent, a baronet, his name Sir Roger de Coverley. His great-grandfather was inventor of that famous country-dance which is called after him. All who know that shire are very well acquainted with the parts and merits of Sir Roger. He is a gentleman that is very singular in his behaviour, but his singularities proceed from his good sense, and are contradictions to the manners of the world, only as he thinks the world is in the wrong. However, this humour creates him no enemies, for he does nothing with sourness or obstinacy, and his being unconfined to modes and forms makes him but the readier and more capable to please and oblige all who know him. When he is in town he lives in Soho Square. It is said he keeps himself a bachelor by reason he was crossed in love by a perverse beautiful widow of the next county to him. Before this disappointment, Sir Roger was what you call a fine gentleman, had often supped with my Lord Rochester and Sir George Etherege, fought a duel upon his first coming to town, and kicked bully Dawson in a public coffee-house for calling him youngster: but being ill used by the above-mentioned widow, he was very serious for a year and a half; and though, his temper being naturally jovial, he at last got over it, he grew careless of himself, and never dressed afterward. He continues to wear a coat and doublet of the same cut that were in fashion at the time of his repulse, which, in his merry humours, he tells us has been in and out twelve times since he first wore it. He is now in his fifty-sixth year, cheerful, gay, and hearty; keeps a good house both in town and country; a great lover of mankind; but there is such a mirthful cast in his behaviour, that he is rather beloved than esteemed.

"His tenants grow rich, his servants look satisfied, all the young women profess love to him, and the young men are glad of his company. When he comes into a house he calls the servants by their names, and talks all the way up stairs to a visit. I must not omit that Sir Roger is a justice of the quorum, that he fills the chair at a quarter-session with great abilities, and three months ago gained universal applause by explaining a passage in the Game Act."

We hear little of Sir Roger, except an occasional opinion, till we reach the 106th number, when Addison takes up the man of whom he said "we are born for each other."

"Having often received an invitation from my friend Sir Roger de Coverley, to pass away a month with him in the country, I last week accompanied him thither, and am settled with him for some time at his country-house, where I intend to form several of my ensuing speculations. Sir Roger, who is very well acquainted with my humour, lets me rise and go to bed when I please, dine at his own table or in my chamber, as I think fit, sit still and say nothing without bidding me be merry. When the gentlemen of the country come to see him, he shows me at a distance. As I have been walking in his fields I have observed them stealing a sight of me over a hedge, and have heard the knight desiring them not to let me see them, for that I hated to be stared at.

"I am the more at ease in Sir Roger's family, because it consists of sober, staid persons; for as the knight is the best master in the world, he seldom changes his servants; and as he is beloved by all about him, his servants never care for leaving him: by this means his domestics are all in years, and grown old with their master. You would take his valet-de-chambre for his brother; his butler is grey-headed, his groom is one of the gravest men that I have ever seen, and his coachman has the looks of a privy-councillor. You see the goodness of the master even in his old house-dog, and in a grey pad that is kept in the stable with great care and tenderness, out of regard to his past services, though he has been useless for several years.

"I could not but observe with a great deal of pleasure the joy that appeared in the countenances of these ancient domestics upon my friend's arrival at his country-seat. Some of them could not refrain from tears at the sight of their old master; every one of them pressed forward to do something for him, and seemed discouraged if they were not employed. At the same time the good old knight, with a mixture of the father and the master of the family, tempered the inquiries after his own affairs with several kind questions relating to themselves. This humanity and goodnature engages everybody to him, so that when he is pleasant upon any of them, all his family are in good humour, and none so much as the person whom he diverts himself with: on the contrary, if he coughs, or betrays any infirmity of old age, it is easy for a stander-by to observe a secret concern in the looks of all his servants.

"My worthy friend has put me under the particular care of his butler, who is a very prudent man, and, as well as the rest of his fellow-servants, wonderfully desirous of pleasing me, because they have often heard their master talk of me as of his particular friend."

Such is the general outline of the character and position of Sir Roger de Coverley. In succeeding numbers we shall present his minuter features.

PORTABLE DIORAMA.—DISSOLVING VIEWS.

IN a former number we gave an outline of the principles on which chiefly depend the effects produced at the Colosseum, the Cosmorama, the Panorama, the Diorama, and other similar exhibitions. Since then we have met with a suggestion by a Mr. Tait of Edinburgh, for the construction of a portable Diorama, which seems worthy of a few further observations.

Mr. Tait communicated to the Society of Arts of Scotland a description of a small apparatus by which the nature and effects of the diorama could be exhibited in an instructing manner. But to understand this, it is necessary to advert to Daguerre's account of

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the mode of painting dioramic pictures, as divulged by him to the French government. A dioramic picture is painted on both sides. It is a large piece of lawn or calico, if possible without a seam, or at least with seams as little perceptible as may be necessary. The colours laid on the *front* of the picture are viewed by reflected light coming from a point above and between the spectator and the picture; while those laid on the *back* of the picture are viewed by transmitted light, emanating from a window behind. In painting the front, the 'lights,' or white tints, are left out, so as to admit the passage of light through the picture from behind: and even in the dark parts no body-colours are used; for though they would show well by reflected light, they would appear as mere black irregular masses by transmitted light. While painting the front, the painter works by reflected light; but while painting the back, by transmitted light; because the effects intended to be produced can only thus be tested.

Generally speaking, when a dioramic scene is represented by day, and then by evening or moonlight, the day effect is painted on the front of the picture, and the night effect on the back; and the admission of light is regulated according as the picture is to be viewed by reflected or transmitted light; in other words, according as it is to be a *picture* or a *transparency*. But in other cases a subject more or less different from the first is represented on the back, by which many of those startling effects have been produced which are so familiar to the visitors at the exhibition in the Regent's Park. The exhibition-room, be it large or small, is provided with shutters, by which the amount of light to be admitted can always be regulated, from broad daylight to total exclusion. If *fog* is to be represented, as it has been in many exhibited dioramas, the picture is placed at a greater or less distance behind a transparent screen; the greater the distance, the more dim and foggy will the scene necessarily appear.

All these arrangements, in order to produce the desired effect to the eye of a spectator, must be so managed that the picture may be at a distance from the eye, in a kind of room or recess; and it is probable that this circumstance led Mr. Tait to the suggestion of a portable diorama. The machine may be a small oblong box, of any dimensions, to be viewed at one end. Small stretching-frames are prepared, over which pieces of transparent paper or linen are stretched to form the pictures. Any one of these, when painted and about to be used, is inserted in a groove in the interior of the box, at a distance equal to two-thirds of the length of the box from the end at which the eye is applied. The eye-hole is not simply a circular or square hole cut in the end of the box, but is a small tube two or three inches in length, placed opposite the 'point of sight' in the picture. The tube projects a little from the box, in order to assist the adjustment of the eye; and the inner end is expanded sufficiently to expose to view the whole of the picture in the box.

As a means of admitting light to act upon both sides of the picture at pleasure, two hinged covers are used, one at the top of the box, and the other at the end remote from the eye. Each cover, by a small pulley and balance weight, or any similar contrivance, is made to remain stationary in any required position. When the top cover is closed and the end one open, light falls on the back, but not on the front of the picture, and a person applying his eye at the tube would see the picture only by transmitted light. When the top cover is open and the ends are closed, the reverse of that occurs, and a spectator views the picture by reflected light. When any medium arrangement is adopted, such as one cover being open and the other partially closed, one closed and the other partially closed, or both partially closed, numerous variations of light and shade and tint in the picture are observed. Passing gleams of sunshine, day melting into night, and this into moonlight—and all similar changes, may be imitated with some approach to completeness.

The inside of the eye-tube, and everything which could distract the eye from the picture, is painted black; while the inner surfaces of the covers which may aid in reflecting light upon the picture are painted white. Screens of fine tissue-paper, Persian silk, or some other thin substance, are placed across the openings when the covers are raised, if a subdued light be required; and remarkable modifications of the effect may be produced by having these media coloured. The pictures may be viewed by the naked eye through the tube, or a lens might be employed to alter the effect.

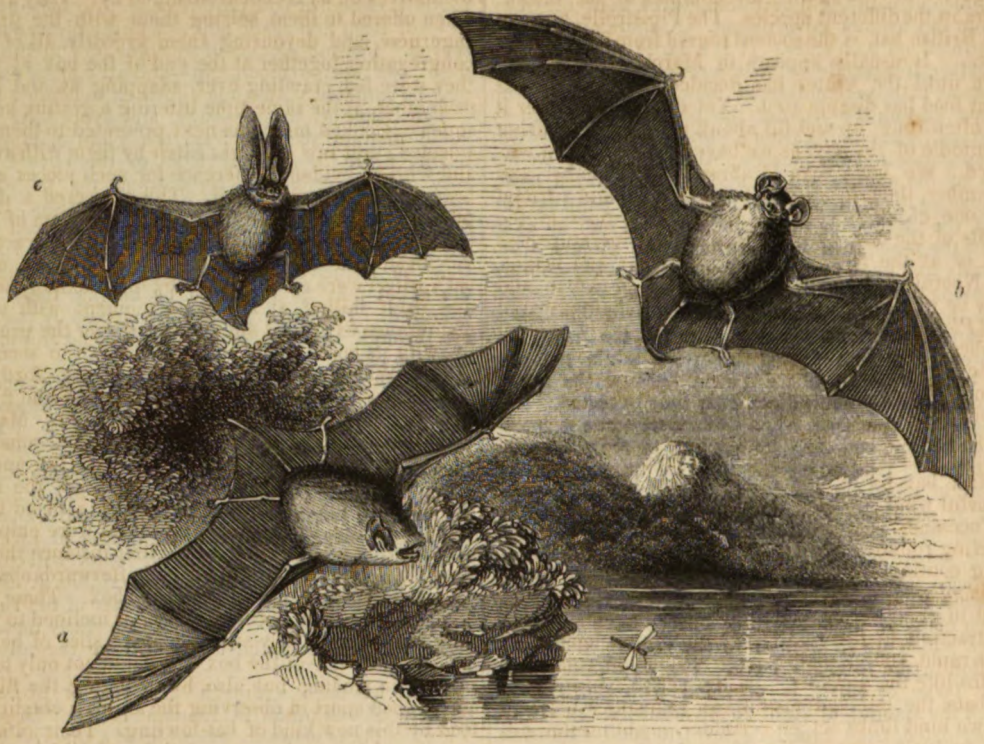
It is not difficult to see that such a contrivance is an exact copy of the large diorama, in all its essential features. The construction of the box is a matter involving no great mechanical difficulties. The painting of the pictures is the feature which calls for most talent; for here attention must be paid to the different character or tone which reflected light and transmitted light throw over a picture, to the degree of opacity or transparency which different pigments will present, to the hues which natural scenery exhibits at different hours of the day, and to the character of the shadows produced by objects. The more carefully these matters are attended to, the better will be the miniature diorama.

There has been, within the last year or two, a kind of pictorial exhibition in London, called "Dissolving Views.' These views are examples of a superior kind of phantasmagoric exhibition, or "magic-lantern," in which striking effects are produced by simple but very ingenious means.

The phenomenon of a "dissolving" view consists in the adjustment of *two* views, or two lantern slides, in such a manner that one shall gradually disappear while the other comes in sight, the images of both occupying the same spot on the screen or wall. It is said that a German named Philipsthal, who introduced the phantasmagoria about sixty years ago, also gave the first rough idea of the "dissolving" views. He was in the habit of representing, among other subjects, the raising of the ghost of Samuel by the Witch of Endor, in which he made the phantom appear to rise from the ground; but he conceived that if he employed two lanterns and slides, making the wick of one rise while he lowered that of the other, and directing both images to one spot, a more aërial and supernatural effect might be produced. This method succeeded, and Philipsthal was led to the adoption of similar arrangements for representing landscape scenery.

The improvements which have been made within the last few years have brought this plan to a point of great excellence. Two sliders or painted glasses are used, illuminated by one intense jet, having their devices represented on a screen, the focalization to one spot being effected by optical means. While one picture is being exhibited, the other is hidden by a cover or shutter; and the effect of "dissolving," which is very remarkable, is produced by the gradual and simultaneous closing of one picture and opening of another. If, while one picture is being exhibited, the other is being changed for a third, and if while this third picture is under exhibition the second be ex changed for a fourth, and so on, an extensive series may be exhibited, each one apparently melting or dissolving into the succeeding one. This, like many other contrivances, appears simple enough when known; but the simplicity does not detract from the merit of the artists who contrived the arrangement.

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[*a*, Common Bat (Vespertilio pipistrellus); *b*, Great Bat (V. noctula); *c*, Long-eared Bat (V. auritus).]

CURIOSITIES OF BRITISH NATURAL HISTORY.

BATS.

It may surprise some of our readers to be informed that sixteen or seventeen distinct species of bats are natives of the British Islands. Of these, however, several are extremely rare, and restricted to certain localities; but some, as the Pipistrelle, or common bat, and the long-eared bat (*Vespertilio auritus*), are everywhere abundant; nor is the great bat (*V. Noctula*) of unfrequent occurrence.

Of all the mammalia the bats alone emulate in their aërial endowments the feathered tenants of the sky; they are essentially flying insectivora. In the air they pass the active periods of their existence, and revel in the exercise of their faculties. Their organs of flight, admirably adapted for their destined purpose, do not consist, as in the bird, of stiff feathers based upon the bones of the fore-arm, but of a membranous expansion stretched over and between the limbs, and to which the bones of the limbs, especially those of the elongated fingers, serve the same purpose as the strips of whalebone in an umbrella. This apparatus can be folded up, and the limbs employed in progression on the ground; on a level surface, however, the bat shuffles awkwardly but quickly along. In the hollows of decayed trees, in the crevices of mouldering masonry, or in rough chinks and fissures, it can crawl and climb about with tolerable rapidity, as also about the wirework of a cage, a circumstance we have often witnessed. It is a smooth and level surface that most embarrasses the bat, but even then it can easily take wing. In the air the bat is all alertness,—it is here that these singular creatures pursue their insect prey—uttering their short sharp cry as they wheel in circling flights, or perform their abrupt and zigzag evolutions. Bats, says White, "drink on the wing like swallows, by sipping the surface as they play over pools and streams. They love to frequent waters, not only for the sake of drinking, but also on account of the insects, which are found over them in the greatest plenty." Often during a warm summer evening have we seen numbers, perhaps several scores, of the common bat (*V. Pipistrellus*) flitting over pools, in chase of gnats and similar insects, or gambolling with each other in a mazy dance, ever and anon uttering sharp shrill cries of exultation and delight; an interesting spectacle to such as love to "trace the woods and lawns and living stream at eve."

The bat is a twilight and nocturnal rambler: it passes the day in its retreat suspended head downwards, clinging to any roughness or projection by the claws of its hinder feet. In this position it hybernates in a state of lethargy; numbers congregating together. Church steeples, hollow trees, old barns, caverns, and similar retreats are its lurking-places; and vast numbers are often found crowded closely together and forming a compact mass. Pennant states that on one occasion, as he was informed by the Rev. Dr. Backhouse, one hundred and eighty-five were taken from under the eaves of Queen's College, Cambridge, and on the next night sixty-three more; all in a torpid condition. They were all of one species, viz., the Noctule, or great bat (*V. Noctula*), the largest of our British bats, measuring fourteen or fifteen inches in the extent of the wings. The great horse-shoe bat haunts the deepest recesses of caverns, where no rays of light can enter. It is found in the caverns at Clifton, and in Kent's Hole near Torquay, a dark and gloomy cavern, where the lesser horse-shoe bat also takes up its abode.

It has been suspected that some of our British bats may possibly migrate, and pass the winter, like the swallow, in some genial region where their insect prey is abundant. For this supposition there is not the slightest foundation: all our bats hybernate; but the period at which they become torpid in their retreats,

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and revive to visit again "the glimpses of the moon," differs in the different species. The Pipistrelle, or common British bat, is the soonest roused from its lethargic trance. It usually appears in March, and does not retire until the winter has decidedly set in, and its insect food has disappeared. Yet during the winter it will often rouse up and flit about, and that too during the middle of the day, as we have ourselves often witnessed. We have seen it abroad in November and December, though the weather was cold, and a friend shot one of these bats just before Christmas in the middle of the day, which, though the temperature was near or at the freezing-point, was clear and bright. The Noctule appears at the latter end of April, and seeks its winter dormitory in August. The long-eared bat (*Plecotus auritus*) is active in the early part of October.

The various species of our bats differ more or less distinctly from each other in the style and character of their flight. The Pipistrelle flits quickly, making abrupt and zigzag turns, and often skims near the ground; the Noctule, which was first noticed as an English bat by White, sweeps high in the air on powerful wings, whence he termed it *altivolans*. On one occasion we saw three or four of this species wheeling round a row of sycamore trees in Kent, uttering continually sharp grating cries. The chafer (*Melolontha vulgaris*) was at the same time flying about in great numbers, and no doubt proved a source of attraction to them. The flight of the long-eared bat is rapid, and it makes large circles, or courses to and fro like the swallow. In the aërial evolutions of the bats, the tail and membrane extending between the two hind limbs act as a rudder, enabling the animals to turn more or less abruptly: it would seem moreover that the tail is to a certain extent a prehensile organ. Mr. Bell, who first noticed the circumstance, observes, that a small portion of the tail in most of our bats is exserted beyond the margin of the interfemoral membrane, and in ascending or descending any rough perpendicular surface this little caudal finger hooks upon such projections as occur, so as to add to the creature's security. When a bat traverses the wires of a cage this action of the tail is particularly conspicuous.

White observes that it is a common notion that bats will descend chimneys "and gnaw men's bacon," and adds that the story is by no means improbable, as a tame bat did not refuse raw flesh, though insects seemed to be most acceptable. The common bat often enters larders, and has been seen clinging to a joint of meat in the act of making a hearty meal upon it. Of this circumstance we are assured by Mr. Bell.

That bats can be tamed is a remarkable fact; but various species differ in the degrees of their docility. Mr. White's bat, a Pipistrelle, was so tame, that it would take flies out of a person's hand. "If you gave it anything to eat it brought its wings round before the mouth, hovering, and hiding its head in the manner of birds of prey when they feed. The adroitness it showed in shearing off the wings of the flies, which were always rejected, was worthy of observation, and pleased me much."

In the 'Proceedings of the Zoological Society' for 1834 we find the following interesting details relative to the habits of the Pipistrelle in captivity, by Mr. G. Daniell. In July, 1833, he received five specimens of this little bat from Elvetham, Hampshire; all were females, and pregnant. "They had been kept in a tin powder-canister for several days, and on being turned loose into a common packing-case with a few strips of deal nailed over it to form a cage, they exhibited much activity, progressing rapidly along the bottom of the box, ascending the bars to the top, and then throwing themselves off as if endeavouring to fly. They ate flies when offered to them, seizing them with the greatest eagerness, and devouring them greedily, all of them congregating together at the end of the box at which they were fed, crawling over, snapping at, and biting each other, at the same time uttering a grating kind of squeak. Cooked meat was next presented to them, and rejected; but raw beef was eaten by them with avidity, and with an evident preference for such pieces as had been moistened with water. This answered a double purpose: the weather being warm, numbers of blue-bottle flies (*Musca vomitoria*, Linn.) were attracted by the meat, and on approaching within range of the bat's wings were struck down by their action, the animal itself falling at the same moment with all its membranes expanded and cowering over the prostrate fly, with its head thrust under, in order to secure its prey. When the head was again drawn forth, the membranes were immediately closed, and the fly was observed to be invariably taken by the head. Mastication appeared to be a laboured occupation, consisting of a succession of eager bites or snaps, the sucking process (if it may be so termed) by which the insect was drawn into the mouth being much assisted by the looseness of the lips. Several minutes were employed in devouring a large fly. In the first instance the flies were eaten entire, but Mr. Daniell afterwards observed detached wings in the bottom of the box. These, however, he never saw rejected, and he is inclined to think that they are generally swallowed. A slice of beef attached to the side of the box was found not only to save trouble in feeding, but also, by attracting the flies, to afford good sport in observing the animals obtain their food by this new kind of bat-fowling. Their olfactory nerves appear to be very acutely sensible. When hanging by their posterior extremities and attached to one of the bars in front of the cage, a small piece of beef at a little distance from their noses would remain unnoticed; but when a fly was placed in the same situation, they would instantly begin snapping at it. The beef they would eat when hungry, but they never refused a fly. In the daytime they often clustered together in a corner, but towards the evening they became very lively, and gave rapid utterance to their harsh grating notes. One of them died on the fifth day after they came into Mr. Daniell's possession, two on the fourteenth, the fourth survived until the eighteenth, and the fifth until the nineteenth day." Each was found to contain a single young one. On the 16th of May, 1834, the same gentleman procured five specimens of the Noctule bat, four females and a male. The latter, which died in two days, was very impatient of confinement, restless and savage, snapping at the females and breaking his teeth in his attempts to escape by biting the wires of the cage. He constantly rejected food. The females were also at first sulky, but in about two days began to eat, preferring small bits of beef in preference to flies, beetles, or gentles. In the course of a few days three of these died, each found to be pregnant with a single offspring. The survivor lived for more than a month, and fed in preference upon the hearts and livers of fowls: she rejected large flies, but partially devoured one or two chafers (*Melolontha vulgaris*). In taking food, it was remarked that the wings were not thrown forward as in the Pipistrelle, the food being seized with an action similar to that of a dog. The water that drained from the food was lapped, but the Noctule did not raise its head in drinking as the Pipistrelle was observed to do. This Noctule took great pains in cleansing herself; she used the hinder limbs as combs, parting the hair on either side from head to tail, and forming a straight line down the middle of the back. The membrane of the wings was cleaned by the creature's nose, which it forced through

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the folds so as to expand them. During her captivity she brought forth a single offspring perfectly destitute of hair and blind: this she wrapped up so closely as to prevent any observation being made. In the evening of the day after giving birth to her offspring she died. But the young one was alive, and attached to the teat of the mother; whence it was removed, wrapped in warm flannel, and fed with milk, which it took from a sponge. It survived eight days, at which time its eyes had not opened, and it had acquired very little hair. The long-eared bat seems to be far more docile than the Noctule. In captivity this elegant species is confident and familiar, very careful in cleaning its fur, and enjoying to gambol and play with others of its species, pretending to bite as we see dogs do when in good-humoured sport. Mr. Bell informs us that Mr. James Sowerby possessed a long-eared bat, which when at liberty in the parlour would come to the hand of those who held a fly towards it, and take the insect without hesitation. "If the insect were held between the lips, the bat would then settle on its young patron's cheek, and take the fly with great gentleness from the mouth; and so far was this familiarity carried, that when either of my young friends made a humming noise with the mouth in imitation of an insect, the bat would search about the lips for the promised dainty."

The Barbastelle (*Vespertilio Barbastellus*, Linn.) is timid and restless, and very impatient of confinement. This bat seems to become torpid more readily than most of our British bats, and also more completely so. The reddish-grey bat (*Vespertilio Nattereri*) was found by Mr. Bell to be very familiar and confiding, readily taking food from the hand; while the whiskered bat (*V. mystacinus*) is timid and restless, and, refusing food, soon dies after its capture. The Barbastelle, the long-eared bat, and the two last mentioned, often hybernate in caverns. Mr. Bell's specimens were found with others in a large chalk cavern in Kent excavated at the bottom of a shaft seventy feet deep.

With regard to the senses possessed by these interesting animals, those of smell and hearing are, as might be expected from the development of their respective organs, wonderfully acute. Connected with the refinement of these senses, we often find, as in the horse-shoe bat, the nose furnished with a membranous foliation of most delicate structure and complex in its arrangement; or, as in the long-eared bat, the external membranous ears largely expanded, having furrows and an inner reduplication, and capable of being folded down. The sight also is quick, and the position of the eyes, which are small, but bright, is favourable for the chase and accurate seizure of insects during rapid flight.

There is a singular property with which the bat is endowed, too remarkable and curious to be passed altogether unnoticed. The wings of these creatures consist, as we have seen, of a delicate and nearly naked membrane of vast amplitude considering the size of the body; but besides this, the nose is in some furnished with a membranous foliation, and in others the external membranous ears are enormously developed. Now these membranous tissues have their sensibility so high, that something like a new sense thereby accrues, as if in aid of that of sight. The modified impressions which the air in quiescence, or in motion, however slight, communicates; the tremulous jar of its currents, its temperature, the indescribable condition of such portions of air as are in contact with different bodies, are all apparently appreciated by the bat. If the eyes of a bat be covered up, nay, if it be even cruelly deprived of sight, it will pursue its course about a room with a thousand obstacles in its way, avoiding them all, neither dashing against a wall nor flying foul of the smallest thing, but threading its way with the utmost precision and quickness, and passing adroitly through apertures, or the interspaces of threads placed purposely across the apartment. This endowment, which almost exceeds belief, has been abundantly demonstrated by the experiments of Spallanzani and others: it is the sense of touch refined to the highest and most exquisite degree of perfection. Thus are the bats aërial in feeling as in habits.

Full, then, of interest is the history of our British Bats, of which we have selected a few details. To watch their ways and actions, what time evening assumes "her gradual dusky veil," when the silence of the tranquil scene is unbroken, save by their sharp reiterated cry, the churr of the goatsucker, and drowsy hum of the shard-borne beetle, is alike pleasing to the contemplative man and the naturalist.

THE PRIVILEGES AND LIABILITIES OF BRITISH SHIPPING.

ACCUSTOMED as we are to the use of articles of foreign produce, and conscious as we may be of the vast maritime arrangements involved in the importation of such articles into England, there are yet probably few, unconnected commercially with the subject, who bestow much thought on the privileges conceded to English shipping, ship-owners, and commanders, in this respect. The tea, the sugar, the hemp, the timber, the wine, which find their way to England, must obviously do so in ships belonging either to British or to foreign ship-owners; and the determination of the ratio in which this freighting privilege shall be divided has led to laws and regulations which merit a little attention.

Mr. M'Culloch states that so long ago as the reign of Henry VII. a law existed whereby the importation of certain commodities was prohibited, unless imported in ships belonging to British owners and manned by British seamen. In the early part of the reign of Elizabeth foreign ships were excluded from our fisheries and coasting-trade. In the time of the Commonwealth foreign ships, belonging to whatever nation, were prohibited from trading with the plantations in America, without having previously obtained a licence. These however were minor regulations, quite eclipsed by a law passed in 1651, which gave a tinge to the maritime transactions of England from that time down to a comparatively recent period. England was at that period in bitter enmity with Holland, whose ship-owners were the great carriers for nearly all the nations of Europe; and it was to crush this power in a rival nation that the republican parliament passed the law in question. By the terms of this enactment, no goods or commodities whatever, grown, produced, or manufactured in Asia, Africa, or America, could be imported into England, Ireland, or the Colonies, except in ships belonging to English subjects, and of which the master and the greater number of the crew were also English. The import-trade of three out of the four quarters of the globe having been thus secured to the English ship-owners, the act proceeded to secure to them as much as possible of the European trade; and for this purpose it declared that no commodities of any European country should be imported into England, except by English ships, or by ships belonging to the countries where the exported goods were produced. This latter clause was intended expressly to act against the Dutch; for scarcely any of their produce came at that time to England, the merchant-ships of Holland having more frequently come to this country in the capacity of carriers for other countries. By the new law, any commodities imported from France, Spain, or Italy, for example, were to be brought either in English ships, or in French, Spanish,

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or Italian ships, as the case might be; thus excluding the carrying ships of Holland.

Shortly afterwards the prohibition was relaxed to this extent: that while Russian and Turkish produce, as well as timber, grain, tar, hemp, flax, wine, spirits, sugar, and a few other articles, were to remain subject to the above regulations, all other commodities might be imported in any ships whatever. But this change was of little importance, for all the most important articles came under the "exceptions." In the reign of Charles II. the national animosity between England and Holland led to an enactment of extreme rigour, carrying the maritime exclusiveness to an extravagant extent; for it prohibited the importation from Holland, the Netherlands, and Germany, of a long list of commodities, under any circumstances, or in any vessels, whether British or foreign, under the penalty of seizure and confiscation of the ships and goods. This last-mentioned act was virtually one of exclusion rather than of commercial regulation; but it had for many years considerable influence on foreign ship-owners.

It was not until a very recent period (1833) that these laws were placed upon such a footing as to allow to foreign ships a privilege at all analogous to that enjoyed by English; and this change was only wrought when experience showed that other nations were about to retaliate. It may be flattering to the national vanity to know that British ships and British seamen are employed to bring foreign produce to our shore; but the maintenance of an analogous principle by other countries would be a perfectly just retaliation. The Americans in 1787, and the Northern powers of Europe at a later period, adopted, or proposed to adopt, measures avowedly copied from the navigation laws of England; so that if timely concessions had not been made, the English ship-owners would have severely suffered.

The regulations which came into force nine years ago, respecting the relative privileges of British and foreign shipping in importing foreign produce into England, involve the following as the chief points:— A list of what are called "enumerated articles" includes those which must be imported under one of these three circumstances: in British ships; in ships of the country where the goods were produced; or in ships of the country from whence the goods were shipped. This list includes masts, timber, boards, tar, tallow, hemp, flax, currants, raisins, figs, prunes, olive oil, corn or grain, wine, brandy, tobacco, wool, shumac, madder, madder-roots, barilla, brimstone, oak-bark, cork, oranges, lemon, linseed, rape-seed, and clover-seed. Goods which are the produce or growth or manufacture of Asia, Africa, or America, are not to be imported into England from any European country, with some few exceptions: this evidently has relation to the employment of English shipping, in preference to foreign European shipping, in bringing produce from distant countries to England. All goods imported from the Channel Islands must freight British ships only. All exports to our own colonies are to be in British ships; as likewise goods carried coastwise from one part of the British Islands to another, or from any one of our colonies to another. Lastly, any goods taken to one of our colonies in a foreign ship must be so taken only in a ship of the country where the goods were produced, or from whence they were exported.

As many advantages are thus given to English shipping over those of foreign countries, it may be asked how these British ships may be always able to designate themselves and to maintain their identity as such. This is effected by a remarkable system of *registration* maintained through the medium of the Custom-House officers. Vessels which are claimed by their owners to be placed on the registry, must be the property of the British sovereign's subjects, and must have been built in the British dominions or dependencies, or have been prize vessels legally condemned. The collectors and comptrollers of the Customs are generally the parties who register the shipping, and who grant certificates of registry to the owners. So severe are the laws in this respect, that if any ship were to exercise the privileges of a British ship before the owners have obtained a certificate of registry, the ship with the whole of its contents would become forfeited to the crown, and might be seized by the officers of the Customs. In order to reduce the immense mass of shipping within something like navigable order, every registered ship is supposed to "belong" to some particular British port, the Customs' officers of which grant the requisite certificate, and make the requisite entry in the register. The port to which a ship is said to belong is generally the nearest one to the residence of the chief owner of the vessel. The proprietorship of every ship, if there be more than one owner, is supposed to be divided into sixty-four equal parts or shares, which may be held by few or many shareholders, not exceeding thirty-two; and not only must every shareholder's name be entered on the certificate of registry, but if any transfer of shares should take place, the registry must be re-effected. No person, with some few exceptions, who has taken the oath of allegiance to a foreign power can become the owner of a British ship.

In order that the registry may be a *bonâ fide* one, it is necessary that the kind and quality of the ship be recorded; and in order to effect this, every ship is thoroughly examined and surveyed before registry by certain Customs' officers and shipwrights, to determine the tonnage and the general character of the ship. The ship is registered by a particular name, which is not to be afterwards changed. If the vessel after being registered undergoes any material alterations, it must be registered anew. If the vessel undergoes repairs in a foreign country exceeding the amount of 1*l.* per ton burden, it ceases to be a British ship, unless the owners or commander can show that such repairs were absolutely necessary at the time for the safe completion of the voyage.

It will thus be perceived that a great many conditions must be fulfilled before a vessel can rank as a British ship, and share in the privileges granted to British shipping. But besides the vessel itself, there are other matters to be attended to before a ship can engage in commerce as a British ship. For instance, every such ship must be navigated during the whole of every voyage, whether with a cargo or in ballast, in every part of the world, by a master who is a British subject, and by a crew of which three-fourths at least are British seamen. If the ship is employed in the coasting trade or in fishing on the British coasts, the whole of the crew must be British seamen. If on any occasion a registered ship is navigated by more than the prescribed number of foreign seamen, a penalty of 10*l.* for each one in excess is incurred.

These regulations render necessary a determination of the question, not only what constitutes a British ship, but who are British seamen? A British seaman, in the legal acceptation of the term, must be a natural-born subject of the British sovereign, or must have been naturalized by act of parliament, or must have been made a denizen, or have become a British subject by the conquest or cession of some newly-acquired territory, or (being a foreigner) must have served on board an English ship of war, in time of war, for the space of three years. Any of these may obtain the privileges, such as they are, of British seamen, and are

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protected by certain laws respecting hiring, payment of wages, and the conduct of their officers.

A committee of the House of Commons, appointed in 1836 to inquire into the causes of the numerous shipwrecks which occurred about that time, suggested, in reference to the registration of ships and seamen, that a 'Mercantile Marine Board,' appointed for the control of merchant-ships generally, should perfect "a system of classification of ships, to the utmost attainable point of accurately defining, by such classification, the real state and condition of every ship registered;" —should "collect information as to the best materials for building, surveying, fitting-out, equipping, loading, and furnishing with the requisite supply of men, provisions, water, and boats, all ships built and registered in the United Kingdom;"—should form certain standards of qualification in seamanship, navigation, and nautical astronomy, to be attained by officers and masters before receiving licences of appointment in the merchant-service;—and should form registry-offices for recording the name, age, capacity, and character of British merchant-seamen, with a view to advance the praiseworthy and set aside the unfitting. These recommendations have not yet been acted on. Meanwhile the "underwriters," or insurers of ships, have adopted a system of registration for their own purposes well worthy of our notice in a future article.

Usefulness of Moles.—Our Correspondent, whose communication on the utility of moles in destroying the wire-worm and other grubs which feed on the plants of the young corn will be found in No. 618, has furnished the following additional information on this subject:—"I had," he says, "a small field of rye-grass and clover, one end of which, early in the spring, was like a honeycomb from workings of moles. A farmer would have destroyed the workers; I, on the contrary, protected them, and not one was destroyed: but I took care to level the mould which they threw up almost every day; and now to the practical result. I lately cut my crop, which was a very good one generally; but at the end, where the moles worked, the crop was better than in any other part; and now not a mole can be discovered in the field. They did the work designed to them by a wise Providence—ate up all the grubs which would have destroyed my young plants, and then took their departure to some neighbour's field, where doubtless they will be trapped. Another remark as regards birds: for example, as to those small birds which are seen upon fruit-trees, such as the titmouse: the vulgar opinion is that they destroy the buds, and thus injure or ruin the crop. Now I never suffer one of that kind of birds to be killed, but rejoice to see them, and protect them; and I would rather see a superabundance of sparrows than none at all, even by way of profit; and the consequence is, that I have very frequently had a crop of fruit when my neighbours have had none. Again, as you pass cottage-gardens, you very frequently see the leaves eaten off the cabbages and gooseberry and currant bushes growing near the doors by caterpillars; whilst cabbages in the fields and fruit-trees at a distance from houses are flourishing and left untouched. Here again the same cause is in operation; the small birds, which would have destroyed the insects, are driven from the doors, but perform their natural operations at a distance from them."

Autumnal Travelling in Siberia.—We made our first journey *en traineau* here; and bad enough it was in that way—on wheels it would have been impossible. The road was very mountainous, and lay through forests for eight or ten versts together, where the snow was drifted to the height of many feet; through which we had to force our way, it not being yet sufficiently hard to resist the horses' feet. In the rapid descents we constantly rolled over and over; and three horses to a light *traineau* had the greatest difficulty in getting up the long steep hills of snow, where there was no solid footing for them. What we should have done with our carriage on such roads we know not; and we had still a long journey before us before we should come to any town where we could leave it till our return from the far East, and to take it on the whole way was out of the question. The next day a council of war was held; when it was decided we should go on to Barnaoul on wheels, a distance of two hundred and eighty versts: but the road was represented as good, and we were told we should find much snow, it being mostly over a dead flat. Accordingly the carriage was fortified with very strong ashen shafts, which were fixed all round it, so as to force a passage through the snow in the case of need; and thus we started for Barnaoul. Bad as our journey had been for some time past, it was evident we had not reached the maximum, and that every day the roads would be worse, till the snow had settled down into solidity, which, in parts where there is little communication, requires some time. We had generally ten or twelve horses the whole of this journey, and did not with all average above five versts an hour. Our first stage was mountainous; but after that the steppes began again, with driving snow and wind, almost amounting to what is called in this country a *buran*, or whirlwind, which is often fatal to travellers if accompanied with snow in any quantity. Having tried the effects of fire, water, and air, under their most fearful forms, we are inclined to give the preeminence in point of horror to the latter. A *buran* which overtakes you in a forest is less formidable, because you cannot well get out of the right track, and the only danger is being buried alive in the snow. But in an open steppe country, when it is very violent, the snow which is falling becomes whirled round, and mixed with that which the wind raises from the ground; so that in broad daylight the driver cannot see an inch before him, and does not know whether he is going to the right or to the left. Many fatal accidents occur in this way; carriages being rolled down precipices, or men and horses frozen to death in the drifted snow, which naturally collects round the only object which interrupts its course for miles and miles.—*Cottrell's Recollections of Siberia.*

Importance of Cities.—If the history of cities and of their influence on their respective territories be deducted from the history of humanity, the narrative remaining would be, as we suspect, of no very attractive description. In such case, the kind of picture which human society must everywhere have presented would be such as we see in the condition, from the earliest time, of the wandering hordes of Mongolians and Tartars, spread over the vast flats of Central Asia. In those regions scarcely anything has been "made" by man. But this most happy circumstance, as it seems to be accounted—this total absence of anything reminding you of human skill and industry—has never been found to realize our poetic ideas of pastoral beauty and innocence. It has called forth enough of the squalid and of the ferocious, but little of the refined, the powerful, or the generous. If anything be certain, it would seem to be certain that man is constituted to realize his destiny from his association with man, more than from any contact with places. The great agency in calling forth his capabilities, whether for good or for evil, is that of his fellows. The picturesque, accordingly, may be with the country, but the intellectual, speaking generally, must be with the town. Agriculture may possess its science, and the farmer, as well as the landowner, may not be devoid of intelligence; but in such connexions, the science and intelligence, in common with the nourishment of the soil, must be derived, in the main, from the studies prosecuted in cities, and from the wealth realized in the traffic of cities. If pasturage is followed by tillage, and if tillage is made to partake of the nature of a study and a science, these signs of improvement are peculiar to lands in which cities make their appearance, and they become progressive only as cities become opulent and powerful.—*Dr. Vaughan's Age of Cities.*

Fossil Trees.—During the progress of the works for reclaiming the extensive waste called White Moss, between Middleton and Failsworth, a large number of trees, of enormous magnitude, have been discovered at a depth of about six feet; some of the oaks have been nearly twelve feet in girth and forty feet in length. Several trees of the oak, fir, and yew tribe have been found to be thoroughly sound, even to the outermost part. Many of the oak-trees have proved more tough and flexible than this tree is under ordinary circumstances. A large quantity of the timber has most unquestionably been on fire. It seems that during some remote age the fossil-trees at White Moss have been burnt, for there are examples of the main shaft of these timbers having been consumed. Singular as it may appear, the trees found in this moss have invariably been met with lying in a direction either south-east or due east.

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[Chandalas.—From 'Les Hindoos' of Solvyn.]

THE CASTES AND TRIBES OF INDIA.

The institution of castes in India is one of the most curious chapters in the social history of mankind. The distinction of ranks and the separation of professions appear to have been established before the remotest era which Hindoo tradition reaches. According to their sacred books the Brahmen proceeded from the mouth of the Creator, which is the seat of wisdom; the Cshatriya from his arm; the Vaisya from his thigh; and the Sudra from his foot. These castes comprise the four orders of a primitive state of society. The Brahmen were priests, the Cshatriyas soldiers, the Vaisyas husbandmen, and the Sudras servants and labourers. The Hindoo religion teaches its followers that it would be impious to confound these different orders. This distinction of caste is the framework of Hindoo society, and all its inconveniences and palpable injustice have been submitted to for ages from a sense of religious duty. The punishment for crime varies in severity with the caste to which the offender belongs, and while the law is merciless towards the Sudra, its force is mitigated when persons of the three higher castes are brought within its reach. In other matters the abuse of natural rights is equally outrageous. For the interest of money on loan the Brahmen only pays two per cent., while three per cent. is exacted from the Cshatriya, four per cent. from the Vaisya, and five per cent. from the Sudra. Mill says:—"As much as the Brahmen is an object of veneration, so much is the Sudra an object of contempt and even of abhorrence to the other classes of his countrymen. The business of the Sudra is servile labour, and their degradation inhuman. Not only is the most abject and grovelling submission imposed upon them as a religious duty, but they are driven from their just and equal share in all the advantages of the social institution." He then cites passages from the sacred books which show that the Sudra was created for the purpose of serving Brahmens; that he was not permitted to accumulate personal property; and that a Brahmen must never read the Veda (the sacred scriptures of the Hindoos) in the presence of Sudras. In the new edition of Mill, by Horace Hayman Wilson, Esq., the Professor of Sanscrit at Oxford, there is the following important note on this passage. Professor Wilson says:—"The law does not justify the term 'abhorrence.' Mr. Mill has collected the extreme texts, and has passed over all the favourable or qualifying passages. The condition of a Sudra in the Hindu system was infinitely preferable to that of the helot, the slave, or the serfs of the Greek, the Roman, and the feudal systems. He was independent; his services were optional: they were not agricultural, but domestic and personal, and claimed adequate compensation. He had the power of accumulating wealth, or injunctions against his so doing would have been superfluous. He had the opportunity of rising to rank, for the Puranas record dynasties of Sudra kings, and even Manu notices their existence. He might study and

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teach religious knowledge, and he might perform religious acts. No doubt the Sudra was considered in some degree the property of the Brahmen, but he had rights, and privileges, and freedom, much beyond any other of the servile classes of antiquity." Mr. Mill himself, in a note elsewhere, observes that "so inconsistent with the laws of human welfare are the institutions described in the ancient Hindu books, that they never *could* have been observed with any accuracy; and when we consider the powerful causes which have operated so long to draw, or rather to force the Hindoos from their inconvenient institutions and customs, the only source of wonder is, that the state of society which they now exhibit should hold so great a resemblance to that which is depicted in their books." In certain cases of necessity the three higher castes were permitted to have recourse for subsistence to the employments of the class or classes below them; but the Sudra, being the lowest, was confined to the species of labour assigned to him, and in seasons of public distress the competition of the Vaisya, or third class, might come to aggravate his previous misery. But, as Professor Wilson points out, he had a resort which the other castes were denied,—emigration; and subsequently the institution of mixed or impure castes threw open their avocations to him. Of these lower castes we must here give a brief notion.

The origin of mixed or impure castes is to be ascribed to the force of circumstances which laws could not prevent. Children were born whose parents belonged to different castes, and they in consequence belonged to no caste, and could not fall into any of the established employments. The infringement of the sacred laws to which they owed their birth rendered them inferior to the degraded Sudra. Charity or plunder could alone furnish them with the means of subsistence. When the number of these outcasts became so great as to render them dangerous to society, the Brahmen, by supernatural means, as the sacred books allege, created a sovereign endowed with the power of arresting the evils of this disordered state. He classified these outcasts, and assigned to each its particular occupation. Instead of plunderers, they became artisans, practised handicrafts, worked in metals, the subdivision of classes being equal to the number of additional occupations which the exigencies of society at the time demanded. This process, whenever it took place, marks the commencement of a new social era. The division of the older society into four classes, comprehending priests, soldiers, husbandmen, and servants, was too simple for a more advanced period. Thirty-six branches of the impure class are mentioned in the sacred books, but the number, as well as the avocation of each, is variously stated by different writers. The lowest caste of all is the offspring of a Sudra with a woman of the sacred caste. This tribe are called Chandalas. Carrying out the corpses of the dead, the execution of criminals, and other degrading and uncleanly employments, are performed by this caste. They are prohibited from living in towns, their very presence being regarded as a pollution; and on meeting a person of a higher caste they are compelled to turn aside lest he should consider himself contaminated by their approach; and yet, while this and other castes are submitting to these indignities and degradations, they are alive to the "pride" rather than to the "shame" of caste. Professor Wilson says:—"The lowest native is no outcast; he has an acknowledged place in society; he is the member of a class; and he is invariably more retentive of the distinction than those above him."

THE WALHALLA.

THE 19th of October, 1842, was a memorable day for Bavaria and its king, for it was that on which was inaugurated a most noble structure, reared for a most noble purpose—to serve as a Pantheon consecrated to genius and intellect—to the heroes, the philosophers, and the poets of universal Germany. Had Ludwig of Bavaria accomplished nothing else, that single edifice would have amply sufficed for his fame, and would have placed his name by the side of those of Pericles and Hadrian, of Leo and the Medici. But when we also call to mind the numerous splendid structures with which he has graced Munich,* rendering it a sanctuary of art, and raising it from comparative obscurity to a very high rank among the capitals of Europe, and that within the short space of twenty years, we have cause to feel astonished; nor is our astonishment altogether unmixed with mortification when we look at home, and perceive that although several handsome buildings have been erected of late years, hardly any of them are of first-rate importance; while some which ought to have been treated as such, and which offered opportunities by far too valuable to be trifled with, have turned out more or less unsatisfactory. No doubt the new houses of parliament will make amends for preceding failures and mishaps in our national edifices, and amply console for them, if consolation it shall be to know that had they been conducted with the same judgment, ability, and zeal, many of our public buildings would have been very greatly superior to what they now are. However, instead of indulging in ungracious comments relative to architectural doings at home, let us proceed to notice what has been done abroad, namely, the Walhalla.

The site of the structure has already been shown in our 274th Number, where the view of it conveys more of an impression of its general effect in combination with the surrounding scenery than of the building, the latter being purposely thrown quite into the distance, so that no more than its general mass is discernible, for the exterior, having no claim to originality, did not call for any minuteness of detail. The structure stands on the north bank of the Danube, so that its principal front, with the flights of steps and terraces leading up to it, faces the south. It is not, however, the mere building or temple taken by itself, but the entire combination produced by the vast constructions on which it is raised that is so exceedingly striking and impressive, and is attended with peculiar grandeur; and had the same building stood upon a flat level, and risen immediately from the ground, the effect would have been altogether different from and inferior to what it now is, when it sits "throned" aloft. Hardly do we know any other edifice, ancient or modern, that has so magnificent an emplacement. Standing at the bottom of the first flight of steps, a person can see only the massive Cyclopean walls of the lower terrace; nor does he obtain a view of the portico until he has reached the steps leading immediately up to it; but when he does come in sight of it, it shows itself to all the greater advantage, bursting upon the eye in towering grandeur, after being lost to it during all the previous approach.

Of the Walhalla itself, the exterior, as we have said, has no pretensions whatever, nor does it affect any, to originality of design; it being in its architecture no more than a repetition of the Parthenon. But if, beautiful as it is, the exterior shows no invention on the part of the architect (Baron von Klenze), widely different is the case with regard to the interior, which

* For an account of some of the buildings, see *Munich*, 'Penny Cyclopædia.'

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is not only most splendid in decoration, but perfectly novel, and as yet quite unique as regards the form of the roof, which corresponds with that of the external one, so that its ends are of pediment shape, and there are intermediate pediments over the horizontal beams resting on the massive piers which divide the plan into three compartments, each of which has a skylight of plate-glass. Thus, while the temple-like character of the exterior is perfectly kept up, and quite free from windows (except one at the north end, corresponding with the door at the other), the interior is lighted much more effectively than would have been the case had the side walls been pierced with such apertures.* Not the least recommendation of all attending the internal form of roof here introduced, is that, while it is altogether original, it is also perfectly consistent and characteristic, since it so completely accords with the outward shape that the spectator is forcibly reminded that the exterior has a pediment at each end. There is no masking—no incongruity—no contradiction of character. You do not find a vaulted Roman hall, within the shell of a Grecian Doric temple. On the other hand, no space is lost for the roof, as would have been had there been a flat coffered ceiling; consequently greater loftiness is obtained. In addition to these advantages, this roof promises the utmost durability, being constructed entirely of cast-iron, but has nevertheless been rendered of most magnificent character within the ceiling or inner surface, being covered with plates of gilt bronze. All the other decorations are of corresponding richness: the pavement is composed entirely of marble, laid in a pattern whose colours are black, white, yellow, and red. The same material of different sorts and hues is employed for the walls, antæ, and columns; nor is the gilding spared, or polychromic embellishment omitted. Corresponding with the richness of the materials employed is that of the design, and all the details. To attempt any description of the latter would be idle, and in regard to the former it must suffice to state that the interior consists of a beautiful Ionic order in columns, and in antæ at the angles of the piers between the compartments; and above that is another or Caryatic order, of colossal female figures representing *Valkyriæ*, or Genii of Walhalla.

In no one respect has cost been spared. The solidity of the construction is quite extraordinary, for not only is the whole entirely of marble, both within and without, but the walls are between eight and nine feet in thickness: it may therefore almost literally be said to be imperishable, calculated to endure for centuries of centuries, and to be a monument that will outlive all but the fame of the illustrious worthies whose busts are deposited within its sanctuary,—all but the fame of Ludwig of Bavaria, who needs no other monument to preserve his name.

A few matter-of-fact particulars may be subjoined, to state the principal dimensions, as given in Bavarian measure, which is something less than our own, the Bavarian foot being to the English one as 0·9517 to 1·000. Extreme length of the plan, including lower flight of steps, three hundred and seventy feet. Greatest breadth, or that of first terrace, two hundred and eighty-six feet. Height of first terrace, sixty-seven feet; height from the ground to level of portico, one hundred and thirty-eight feet; height to the apex of pediment, one hundred and ninety-five feet. Temple, measured at base of columns, ninety-eight by two hundred and seventy-five feet: interior, one hundred and fifty by fifty-seven feet; or total length, including the farther compartment at the north end, behind the screen of columns, one hundred and seventy-eight feet. Beneath the temple are massive substructures of vaulted chambers, entered from a door on the first terrace, and forming an ascending plane from that level.

* How few and comparatively small apertures are required for lighting an apartment, when they are made in the ceiling, instead of the sides, is strikingly manifested in the large room of the General Commercial Hall, Threadneedle Street, which is now beautifully lighted, although there would have been little more than darkness visible, had the apertures which are now skylights been made side-windows.

Pekin.—A Russian officer, M. Kovenko, has published in the 'Annuaire des Mines de Russie,' a sketch of environs of Pekin, some extracts from which may interest our readers at the present moment. For a century past, Russia has maintained a convent and school at Pekin, where her interpreters receive their education in Chinese and Mantchou. Every ten years the members of these two establishments are changed, and fresh monks and pupils are sent from St. Petersburg. During their stay at Pekin, the Russians are free to see all things and visit all places without awakening the restless jealousy of the government. Pekin, according to M. Kovenko, is situated in a plain bounded to the north-west by a series of mountains which the Chinese divide into northern and western, according to their position with reference to the city. The northern mountains are a day's journey from Pekin; that being no great distance, for the Chinese never travel more than five and twenty of our miles in a day. This road in summer is very picturesque, and the country highly cultivated. The yellow millet is the Chinese peasant's plant, *par excellence*. Its grain is the basis of his nutriment; the stalk is food for his cattle, in the place of hay, which they have never thought of cutting. The straw of another species of millet, which attains a height of fifteen feet, is used to make the fences of gardens, and serves also for fuel. Near these northern mountains are some springs, having a temperature of forty-five degrees. The water is conducted by pipes into baths cut in the calcareous rock, and lined with sheets of lead. Early in the spring crowds assemble at this spot in search of health or for the mere pleasures of the promenade. The Imperial family has a palace here, and there are several temples in the neighbourhood. In these temples it is that the weary traveller may seek repose; but the hospitality of the priests of Khé-san and of Da-o is by no means gratuitous. M. Kovenko asserts that a few hours' rest will cost about eighteen roubles (between 16*s.* and 17*s.*), and upwards of twenty-five roubles are often paid for a day's. A multitude of fruit-trees grow in the valleys of these mountains, as well as willows, firs, juniper-trees, and cypresses, but these do not form forests of any considerable extent. The western mountains are remarkable for the coal which they enclose. So abundant is it, that a space of half a league cannot be traversed without meeting with rich strata. Yet, either because of this very abundance, or from the inveterate habit which the Chinese have of leaving all things unperfected, the art of mining is yet in its infancy amongst them. Machinery to lighten labour is there unknown. They have not even an idea of the pumps indispensable to draw off the water. If local circumstances allow, they cut drainage-galleries; if not, they abandon the working when the inundation has gained too far upon them. Their system of ventilation consists in making openings at certain distances, over which they place wheels turned by men. But these wheels, though incessantly in motion, introduce very little air into the mines. The mattock, pick-axe, and hammer are the mining instruments. A furrow is traced with the pick-axe, the mattock is inserted and driven in with the hammer; and in this manner lumps of coal are detached, weighing from sixty to eighty pounds. Coal is at a moderate price in the capital. It is burnt in bronze vases, or its heat is distributed along the wall by means of pipes. These precautions against cold are very necessary at Pekin, and not the mere consequences of that strange habit which makes the Chinese heat all their drinks, even their wine. It freezes and snows often, and on the 31st of December, 1820, M. Timkowski found the thermometer there down to twelve degrees below zero.—*Athenæum.*

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[Hatfield House.]

PROGRESSES OF QUEEN ELIZABETH.

FROM HATFIELD PREVIOUS TO HER CORONATION.

AMONG the many alterations which increased facility of intercommunication has produced, is one that may not perhaps at first occur to us—this is the lessening of processional pomp and magnificence. When the removal of a nobleman's or a gentleman's family from one of his residences to another involved the transport also of much of his household stuff, and when neither roads nor vehicles admitted of rapid movement, such removals were unfrequent, the train naturally assumed the processional form, the rarity gave it the character of a show, and the occasion, the farewell or the welcoming of the local chief, gave it that of a holiday. What arose from necessity became consecrated by custom, and ultimately elevated by art into a gorgeous though sometimes rude display of pomp. The rank and dignity of the individual were considered to be involved in the number and magnificence of his attendant retinue, and his popularity or political influence was indicated by the reception he met with in the places through which he passed. On the embassy of Becket to France in 1158, he was attended by two hundred knights, besides barons and nobles, a host of domestics, eight covered waggons, each guarded by armed men and a fierce dog, containing his kitchen and bedchamber furniture, that of his chapel, his plate, his wardrobe, in which he had twenty-four changes of apparel, his hawks, hounds, huntsmen, &c., with twelve sumpter horses, each ridden by a monkey; and two hundred and fifty boys, who preceded the train on entering a town, singing national songs. In a later reign, the magnificence of Wolsey was not less remarkable, though the style was somewhat altered. At the present time, when the queen and court travel by railroad at the rate of forty miles an hour, or in post-chariots at fifteen—when judges go their circuits by similar conveyances, nothing of the old custom remains to us, except the heavy pomp of funereal processions, and the scarcely less heavy and unpoetical exhibition offered to the citizens of London on Lord Mayor's day, or, occasionally, the less pompous but more impressive ceremony of the opening of parliament by the sovereign in person.

During a period when it was a work of great labour, requiring much time, and occasioning enormous expense for subjects, particularly those from the remoter districts, to visit the court and look upon their sovereign, it became a practice with all such monarchs as thought they deserved or wished to acquire popularity, to make Progresses through the different parts of their territories. As the necessities for the long and cumbrous trains became less imperative, efforts were made to give these exhibitions more of an ornamental and intellectual character, though frequently of a formal and pedantic description, on the part alike of visitors and visited. It was during the reigns of Queen Elizabeth and King James I. that these entertainments reached their highest elevation; and from that of the last we may date their extinction: so vain are the efforts of art to prolong the existence of any state of manners not in unison with the more material conveniences and improvements of the time. As a record, however, of a state of manners which can never return, and affording also occasion of exhibiting specimens of the current literature, we purpose giving a few papers upon the Progresses of Queen Elizabeth, abundant materials for which are found in the three bulky quartos under that title, published by the late John Nichols, Esq., though we shall not confine ourselves to this single authority.

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The princess Elizabeth, as is generally known, passed the last part of the reign of Queen Mary in a sort of half confinement in the then royal palace of Hatfield, now the seat of the Marquis of Salisbury, to whose good taste the public are indebted for the preservation of the building in all its substantial features as it then existed. It was here that, on the 18th of November, 1558 (Queen Mary having died early in the morning of the 17th), several lords of the Privy Council waited on her to announce her accession to the throne. She remained here till the 23rd, when she began her first Progress, which we may say only terminated with her Coronation.

At her departure she was attended by more than a thousand persons. At Highgate she was met by the bishops, and at the foot of Highgate Hill by the lord mayor and corporation of London, by whom she was accompanied to the Charter-House, then the residence of Lord North. On the 28th she proceeded to the Tower. "All the streets she was to pass, even to the Tower, were new gravelled; and so she rid through Barbican and Cripplegate, and along London Wall unto Bishopsgate, and thence up to Leadenhall, and so through Grasschurch Street and Fanchurch Street, turning down Mark Lane into Tower Street, and so to the Tower." How pleasantly these "old familiar" names fall upon the ear, speaking of so little change, that they are yet the perfect direction of the road from the Charter-House to the Tower! "Before her rode many gentlemen, knights, and nobles: after them came the trumpeters blowing; then all the heralds in array; my lord mayor, holding the queen's sceptre, riding with garter; my lord of Pembroke bare the queen's sword. Then came her grace on horseback, apparelled in purple velvet, with a scarf about her neck, the sergeants-of-arms being about her person. Next after her rode Sir Robert Dudley (afterwards Earl of Leicester), master of her horse; and so the guard with halberds. There was great shooting of guns; the like was never heard before. In certain places stood children, who made speeches to her as she passed, and in other places was singing and playing with regals." At the Tower she remained till Dec. 5, on which day she went by water to Somerset Place, "trumpets sounding, much melody accompanying, and universal expressions of joy among the people." The ceremonies attendant on the funeral of Mary occupied a few days, and on the 23rd Elizabeth left Somerset House for her palace at Westminster, where she kept her Christmas. On Thursday, Jan. 12, 1558-9, she left Westminster, to go by water to the Tower, "the lord mayor and aldermen in their barge, and all the citizens, with their barges decked and trimmed with targets and banners of their mysteries, accordingly attending on her grace. The batchelor's barge of the lord mayor's company, to wit, the Mercers', had their barge with a foist trimmed with three tops, and artillery aboard, gallantly appointed to wait upon them, shooting off lustily as they went, with great and pleasant melody of instruments, which played in most sweet and heavenly manner. Her grace shot the bridge about two of the clock in the afternoon, at the still of the ebb, the lord mayor and the rest following her barge, attending the same till her Majesty took land at the Privy Stairs of the Tower Wharf."

On the 14th commenced the grand display of her progress by land from the Tower to Westminster, previous to her coronation. A detailed description of this 'passage' exists in a curious tract, entitled 'The Passage of our most dread Sovereign Lady Queen Elizabeth through the City of London to Westminster the day before her Coronation,' anno 1558-9, published on the 23rd of the same month. In this it is stated that about two o'clock of the afternoon she "marched from the Tower to pass through the city of London towards Westminster, richly furnished and most honourably accompanied, as well with gentlemen, barons, and other the nobility of the realm, as also with a notable train of goodly and beautiful ladies, richly appointed; and entering the city, was of the people received marvellous entirely, as appeared by the assembly, prayers, wishes, welcomings, cries, tender words, and all other signs, which argue a wonderful earnest love of most obedient subjects towards their sovereign; and on the other side her grace, by holding up her hands and merry countenance to such as stood far off, and most tender and gentle language to those that stood near to her grace, did declare herself no less thankfully to receive her people's good will than they lovingly offered it unto her. To all that wished her grace well she gave hearty thanks, and to such as asked God to save her grace, she said again God save them all, and she thanked them with all her heart; so that on either side there was nothing but gladness, nothing but prayer, nothing but comfort. The Queen's majesty rejoiced marvellously to see that so exceedingly showed towards her grace which all good princes have ever desired; I mean, so earnest love of subjects, so evidently declared even to her grace's own person, being carried in the midst of them. The people again were wonderfully ravished with the loving answers and gestures of their princess, like to which they had before tried at her first coming to the Tower from Hatfield. This her grace's loving behaviour preconceived in the people's heads, upon these considerations, was then thoroughly confirmed, and indeed implanted a wonderful hope in them touching her worthy government in the rest of her reign. For in all her passage she did not only show her most gracious love toward the people in general, but also privately, if the baser personages had offered her grace any flowers or such like as a signification of their good will, or moved to her any suit, she most gently, to the common rejoicing of all lookers on and private comfort of the party, staid her chariot and heard their requests.

* * * * *

"Near unto Fanchurch was erected a scaffold richly furnished, whereon stood a noise of instruments,* and a child in costly apparel, which was appointed to welcome the Queen's majesty in the whole city's behalf. Against which place, when her grace came, of her own will she commanded the chariot to be stopped, and that the noise might be appeased until the child had uttered his welcoming oration, which he spoke in English metre, as here followeth:—

"Oh! peerless sovereign Queen, behold what this thy town
Hath thee presented with, at thy first entrance here;
Behold with how rich hope she leadeth thee to thy crown,
Behold with what two gifts she comforteth thy cheer!

The first is blessings' tongues, which many a welcome say,
Which pray thou mayst do well, which praise thee to the sky;
Which wish to thee long life, which bless this happy day.
Which to thy kingdom heaps all that in tongues can lie.

The second is true hearts, which love thee from their roots,
Whose suit is triumph now, and ruleth all the game:
Whose faithfulness have won, and all untruth driven out;
Which skip for joy when as they hear thy happy name.

Welcome therefore, O Queen, as much as heart can think,
Welcome again, O Queen, as much as tongue can tell;
Welcome to joyous tongues, and hearts that will not shrink:
God thee preserve, we pray, and wish thee ever well.

"At which words of the last line the whole people gave a great shout, wishing with one assent, as the child had said. And the Queen's majesty thanked most heartily both the city for this her gentle receiving at the first, and also the people for confirming the same.

* A *noise* is a band—a noise of music, a band of music.

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Here was noted in the Queen's countenance, during the time the child spake, besides a perpetual attentiveness in her face, a marvellous change in look, as the child's words touched either her person or the people's tongues or hearts. So that she, with rejoicing visage, did evidently declare that the words took no less place in her mind, than they were most heartily pronounced by the child, as from all the hearts of her most hearty citizens."

As well as being spoken, copies of all the addresses were stuck up in convenient places, in Latin as well as English. In Gracechurch Street, opposite the "sign of the Eagle," a "gorgeous and sumptuous ark" was erected across the street, elevated in three stages above the three arches, which gave a passage to the procession and its spectators. In the lowest stage, upon one throne sat richly dressed representatives of Henry VII. and Elizabeth, the one enclosed in a red rose, the other in a white rose, and both royally crowned, with their hands joined. From the two roses sprang two branches uniting into one toward the second stage, and in this was placed a figure of Henry VIII., crowned, and seated by the side of Queen Anne Boleyn. From their seat ascended one branch to the highest stage, in which was a figure of Elizabeth herself. All the figures were represented by children, and in front was a standing for one, whose task it was to declare in verse the "whole meaning of the said pageant," the blessings of unity and the cessation of civil wars, to her majesty, with which she declared herself well pleased.

"Ere the Queen's majesty came within hearing of this pageant, she sent certain [persons], as also at all the other pageants, to require the people to be silent, for her Majesty was disposed to hear all that should be said unto her." A curious specimen of the orderliness of the populace, if the desire was complied with. On advancing through Cornhill, the Conduit "was curiously trimmed against that time with rich banners adorned, and a noise of loud instruments on the top thereof." The second pageant, at the "nether end of Cornhill," had for its title "The Seat of Worthy Governance;" children here again representing the characters of the Queen with four allegorical attendants, "Pure Religion, Love of Subjects, Wisdom, and Justice; which did tread their contrary vices under their feet; that is, to wit, Pure Religion did tread upon Superstition and Ignorance; Love of Subjects did tread upon Rebellion and Insolence; Wisdom did tread upon Folly and Vain Glory; Justice did tread upon Adulation and Bribery." Each was distinctively marked, not only by "their names set in plain and perfect writing on their breasts," but by their apparel; and one child was appointed to deliver the versified rather than poetical explanation. The Queen on all these occasions stayed her chariot, most attentively listened to the addresses, and thanked the city in appropriate terms for their pains.

"The Great Conduit in Cheap was also ornamented; and against Soper's Lane was another pageant of eight children representing the Eight Beatitudes, who also delivered an address; but the most marked one was that at the Little Conduit in Cheap, where the acceptance by the Queen of the Bible, and the promise oftentimes to read it, must have been highly satisfactory to the spectators in the then state of religious feeling in the city.

"Soon after that her grace passed the Cross she had espied the pageant erected at the Little Conduit in Cheap, and incontinent required to know what it might signify. And it was told her grace that there was placed Time. 'Time!' quoth she, 'and Time hath brought me hither.' And so forth the whole matter was opened to her grace, as hereafter shall be declared in the description of the pageant. But in the opening, when her grace understood that the Bible in English should be delivered unto her by Truth, which was therein represented by a child, she thanked the City for that gift, and said that she would oftentimes read over that book, commanding Sir John Parrat, one of the knights which held up her canopy, to go before and to receive the book. But learning that it should be delivered unto her grace down by a silken lace, she caused him to stay, and so passed forward till she came against the aldermen in the high end of Cheap tofore the Little Conduit, where the companies of the City ended which began at Fanchurch, and stood along the streets, one by another, enclosed with rails hanged with cloth, and themselves well apparelled with many rich furs, and their livery-hoods upon their shoulders, in comely and seemly manner, having before them sundry persons well apparelled in silks and chains of gold, as whifflers and guarders of the said companies, besides a number of rich hangings, as well of tapestry, arras, cloths of gold, silver, velvet, damask, satin, and other silks plentifully hanged all the way as the Queen's highness passed from the Tower through the City. Out at the windows and the penthouses of every house did hang a number of rich and costly banners and streamers till her grace came to the upper end of Cheap. And there, by appointment, the Right Worshipful Master Ranulph Cholmely, Recorder of the City, presented to the Queen's majesty a purse of crimson satin, richly wrought with gold, wherein the City gave unto the Queen's majesty a thousand marks in gold, as Master Recorder did declare briefly unto the Queen's majesty. . . . The Queen's majesty, with both her hands, took the purse, and answered him again marvellous pithily.

* * * * * *

"And in the same pageant was advanced two hills, or mountains, of convenient height. The one of them, being on the north side of the same pageant, was made cragged, barren, and stony; in the which was erected one tree artificially made, all withered and dead, with branches accordingly. And under the same tree, at the foot thereof, sate one in homely and rude apparel, crookedly, and in a mourning manner, having over his head, in a table, written in Latin and English, his name, which was 'Ruinosa Respublica'—'A decayed Commonweal.' And upon the same withered tree were fixed certain tables, wherein were written proper sentences expressing the causes of the decay of 'Commonweal.' The other hill, on the south side, was made fair, fresh, green, and beautiful,—the ground thereof full of flowers and beauty; and on the same was erected also one tree very fresh and fair, and under the which stood upright one fresh personage, well apparelled and appointed, whose name also was written both in English and Latin, which was 'Respublica bene instituta'—'A flourishing Commonweal.' And upon the same trees also were fixed certain tables, containing sentences which expressed the causes of a flourishing commonweal. In the middle, between the said hills, was made artificially one hollow place or cave, with door and lock enclosed; out of which, a little before the Queen's highness coming thither, issued one personage whose name was Time, apparelled like an old man, with a scythe in his hand, having wings artificially made, leading a personage of lesser stature than himself, which was finely and well apparelled, all clad in white silk; and directly over her head was set her name and title, in Latin and English, 'Temporis filia'—'The daughter of Time.' Which two so appointed went forward toward the south side of the pageant. And on her breast was written her proper name, which was 'Veritas'—'Truth,' who held a book in her hand, upon the which was written 'Ver-

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bum Veritatis'—the 'Word of Truth.' And out of the south side of the pageant was cast a standing for a child, which should interpret the said pageant,' which was accordingly done in English and Latin.

The Queen, on receiving the book, kissed it, and laid it to her breast. At St. Paul's the scholars welcomed her with a Latin oration in prose, and some Latin verses, to which she listened attentively, and graciously received the written copies. At Ludgate the gate was decorated and a "noise of instruments" placed. At the conduit in Fleet Street was erected the fifth and last pageant, in which a "seemly and meet personage" represented "Deborah the judge and restorer of Israel," with six other personages, two denoting the nobility, two the clergy, and two the commonalty; while a child again addressed her in verses allusive to the pageant. At St. Dunstan's the "children of the hospital*" were drawn up, and one addressed her in a Latin oration. "From thence her grace came to Temple Bar, which was dressed finely with the two images of Gotmagot the Albion and Corineus the Briton, two giants big in stature, furnished accordingly; which held in their hands, even above the gate, a tablet, wherein was written, in Latin verses, the effect of all the pageants, which the City before had erected." In a smaller tablet was the same in English as follows:—

"Behold here in one view thou mayst see all that plain,
O Princess, to this thy people the only stay:
What each where thou hast seen in this wide town, again
This one arch whatsoever the rest contained doth say.

The first arch as true heir unto thy father dear,
Did set thee in the throne where thy grandfather sat;
The second did confirm thy seat as Princess here,
Virtues now bearing sway, and Vices beat down flat;

The third, if that thou would'st go on as thou began,
Declared thee to be blessed on every side;
The fourth did open Truth, and also taught thee when
The Commonweal stood well, and when it did thence slide;

The fifth as Deborah declared thee to be sent
From Heaven, a long comfort to us thy subjects all:
Therefore go on, O Queen, on whom our hope is bent,
And take with thee this wish of thy town as finall.

Live long, and as long reign, adorning thy countrie
With Virtues, and maintain thy people's hope of thee;
For thus, thus Heaven is won; thus must you pierce the sky:
This is by Virtue wrought, all other must needs die."

And so with a farewell address from a child on the south side "her grace departed forth through Temple Bar towards Westminster," with shouting and crying of the people, and the firing of the Tower ordnance.

* Christ Church.

INSURANCE OF SHIPPING.—"LLOYD'S LIST."

MANY a newspaper reader has probably marvelled as to the nature and meaning of "Lloyd's Coffee-house." He may have seen, that in all lists or catalogues of shipping, entering or leaving foreign ports and harbours, Lloyd's agents or correspondents appear to furnish the information; and not unfrequently a letter appears from the Admiralty, giving information respecting lighthouses, beacons, and other matters pertaining to marine affairs, addressed to the Secretary at Lloyd's.

Who is this Lloyd? it may be asked, and what kind of a Coffee-house does he keep? And what has his Coffee-house to do with shipping affairs? And why does Lloyd employ so many agents and correspondents, and what advantage does he reap by so doing? The apparent incongruity of the matter may be cleared up at once by saying, that there is a Society or Committee of mercantile men, meeting occasionally for a defined object, who originally held their meetings in a subscription-room attached to a Coffee-house, at or near the Royal Exchange. "Lloyd's Coffee-house" hence became generally known in connection with the objects of that Committee; and in accordance with the brevity of commercial language, the simple word "Lloyd's" came to imply the Committee itself, and the general object for which it was formed; and the name has adhered to it ever since, though neither Lloyd nor the Coffee-house has had anything to do in the matter for some years.

The subscribers to Lloyd's are principally a number of persons known by the appellation of "Underwriters," who insure shipping on the same principle as a Fire-office insures buildings. But it is a remarkable feature, that these underwriters effect insurances individually, instead of combining their funds into a joint-stock for the purpose. The rooms which they engage (once at the Coffee-house over the Royal Exchange, from whence the system is named, and now, during the rebuilding of the Exchange, at the South Sea-house) serve as a kind of bazaar or general office, convenient for transacting business between the insurers and the insured.

To understand the origin of this system, we must look back to the last century. The present "Lloyd's Committee" arose out of the amalgamation of two societies, one of which had existed from 1760 to 1834; and the other from 1798 to 1834; the former a society of underwriters, and the latter a society of shipowners. The shipowners' society took cognizance of all matters relating to the general interests of shipowners; while the underwriters' society was established for the purpose of preparing and publishing annually a Register of British merchant shipping, in which the age, burden, limit, quality, and condition of vessels were so accurately entered, as in a great measure to guide the merchant, the shipowner, and the underwriter in their proceedings. About eight or nine years ago, circumstances occurred which led to a change in the system, by bringing merchants, shipowners, and underwriters all into one society, for the general benefit of all. Thus arose the present "Lloyd's" association, of which we will first describe the general constitution, and then the mode of working.

Lloyd's Register Society is composed of subscribers, of whom twenty-four are chosen to form a managing committee, viz. eight merchants, eight shipowners, and eight underwriters. Two of each class go out of office annually, and are replaced by others; and the election is so managed as to give shipowners and underwriters equal power in the Society. This Committee, as in analogous cases, has power to appoint the various servants of the Society, and to manage the general affairs.

The object of the society is based on the principle of assigning to merchant-ships, wheresoever built, or belonging to any one who may choose to co-operate with the society, a character that shall indicate as nearly as possible their real and intrinsic quality. The advantages are threefold. First, a merchant who freights a vessel with goods to a foreign country can form some opinion of the soundness of the vessel, if it occupies a place in Lloyd's Register; secondly, a person who is about to buy a ship, or a share in a ship, has something besides his own penetration to depend upon, when the ship is classed in the Register; and thirdly, an underwriter who insures a ship before it sets out on a voyage, can form a judgment as to the premium which he must charge, from the place which the ship occupies on the Register. The preparation of this Register thus becomes a matter of paramount importance, and engages the closest attention of the Committee. At one time the classification was effected on a very imperfect principle. Instead of classing the ships according to the actual state and condition ascer-

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tained by a careful surveyor, the character of a ship was stamped wholly by her age and the place where she was built, without any regard to the manner in which she was constructed, the wear or damage she might have sustained, or the repairs she might from time to time have received. This had a lamentable influence on the mode of constructing shipping; for, to use the words of a Parliamentary Committee on the subject—"All new vessels, however slightly constructed, were entitled to be registered in the first class for a given number of years, varying from six to twelve, after which the strongest ships were placed on a level with the weakest, being excluded from the first class when the prescribed period of years had expired." Now, however, the classification is effected according to the actual merits of the ship. Surveyors are appointed by the committee at all the principal ports in the kingdom, to examine in the most scrupulous manner every ship which is to be placed on the Register. It is optional with a ship-owner whether he will or will not have his ship entered in this Register; but if he does, the surveyor examines it, and charges a certain fee for the registration. The surveyor at any particular port transmits periodically to London an account of all ships lost, broken up, or otherwise destroyed, belonging to his district; and also of all vessels building, the state of their progress, the changes which may have occurred in the ownership, and indeed every fact calculated to elucidate the actual state of the shipping at any given period. All the surveyors have been either practical nautical men or practical shipwrights, who have given up all other occupations to attend to the interests of Lloyd's Society.

It was stated in 1836 that out of twelve thousand British merchant-vessels above fifty tons burthen, seven thousand were entered in Lloyd's Register. All these entries were made in a book which is reprinted once a year. The fees for examining and classifying a ship vary from half a guinea to three guineas. When a surveyor has examined a new ship, he transmits to London full particulars of the examination, and mentions the class in which he thinks the ship ought to be placed in the Register. The Committee, if they think the reasons are good, assent to that recommendation; but if their inference from facts be different from his, they determine the class in which to place the ship. If a shipowner is so fortunate as to have his ship classed "A. 1," he seldom fails to append that character to the name of the ship in all his advertisements, as an honourable testimony to the character of the vessel.

Ships are seldom or never insured unless they are registered in Lloyd's list, for the underwriters and insurance societies have then no adequate means of knowing the character of the vessel. Nor can a shipowner sell a vessel so readily if it be not registered in the same list. Merchants, shipowners, and underwriters all feel confidence in this register, and hence its great importance.

There is a committee whose object is to superintend the preparation of the register above alluded to; and also another for the management of the underwriters' proceedings generally. Thus, agents are appointed in all the principal ports of the world, who transmit regularly accounts of the departures from and arrivals at their ports, as well as of losses and other casualties; and, in general, all such information as may be supposed of importance in guiding the judgments of the underwriters. There is an open subscription-room at "Lloyd's," in which all these items of information are entered in books, open to the perusal of the subscribers. It is from the register of shipping that the underwriter forms an opinion of the character of the vessels which he may insure; and from the daily list of occurrences at sea he learns the fate of these several vessels.

Some of the merchants and shipowners manage their own insurance transactions, that is, deal immediately with the underwriters; but there are others who employ insurance-brokers, paying a small per-centage for their services. The sum paid for insuring a given vessel at a given time depends on a large number of circumstances, such as the age of the vessel, its size and condition, the character of its captain or master for skill and steadiness, the nature of the cargo, the length of the voyage on which the vessel is destined, and the average amount of danger on that route. Nothing but the greatest experience can determine the probability of loss under these combined circumstances; but the effect of competition is to cause all these points to be thoroughly investigated.

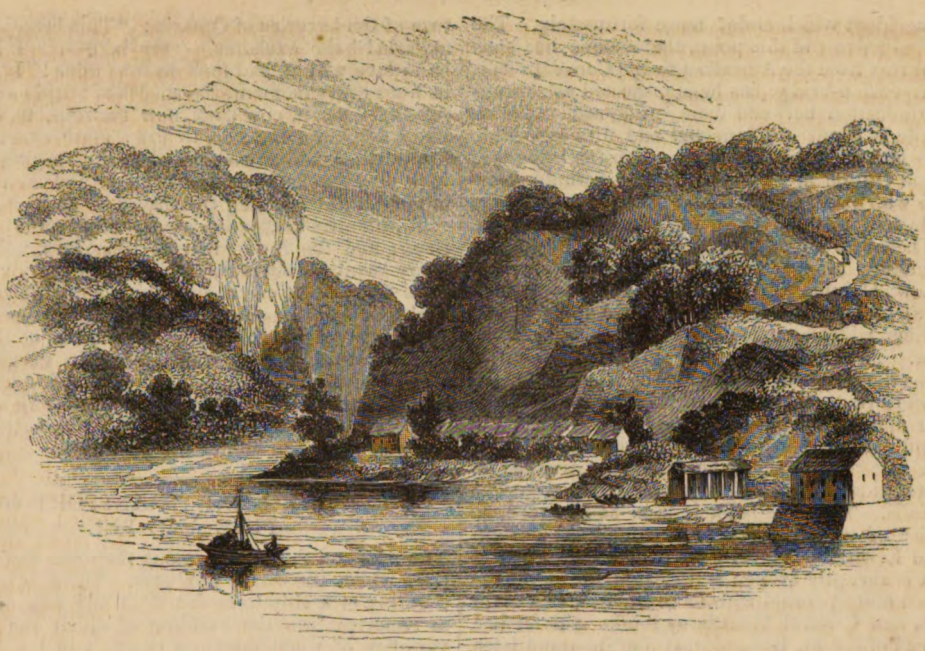
Marine insurance is effected on three different systems; by individuals, by clubs, and by companies. The individual insurers are the underwriters of whom we have spoken, each of whom speculates on his own account. The clubs are associations of shipowners, who agree to divide among themselves the losses sustained in respect of any one of their ships. They enter their ships according to the estimated value, satisfactorily to all the members of the club; and in the event of any loss, the amount is collected individually from all the members, each paying a prescribed quota.

With respect to companies, none such were permitted by law to effect marine insurances until 1824, except two chartered companies; but since that period two others have been added. Mr. M'Culloch stated, in the second edition of his Dictionary, that about one-fifth of the marine insurance transactions is in the hands of the London companies, and four-fifths in the hands of underwriters and of country insurers.

In the policy of a marine insurance the general nature of the proposed voyage is set forth; and the insurance itself becomes void if the ship is lost or damaged under certain circumstances. Thus, acts of our own government, breaches of the revenue laws, breaches of the law of nations, deviations of route, protraction of the voyage,—all are, under certain restrictions, deemed to exonerate the underwriter or company from bearing the losses of a shipowner. The object in view is essentially to secure the shipowner from loss arising out of the "perils of the sea."

The Arein.—The Jaman is sometimes in the winter and spring a dangerous passage, as well on account of the depth of the snow, as in being subject to avalanches, and to the peculiar *tourmente*, as the mountaineers expressively term the snowy winds or windy snows, called the *arein*, a word which signifies in the patois of the country a sandy snow, the particles thereof being dry and brittle. These *areins* are formed by one layer of snow falling upon another, already frozen and hard, and a strong wind forcing its way between the two, slicing off, if I may be allowed so homely an expression, the latest fallen and uppermost, and driving it down the inclined and icy plain on which it has sought its short repose, with a fury that sweeps before it trees, châlets, herds, human beings, all in one bewildering, blinding hurricane, condemning the unfortunate passenger to certain death. In 1767 one of these *areins* swept away, between the Jaman and the village of Allières in Fribourg, on which we were now looking down in all the serenity of a summer's day, a number of large firs and several houses, which it carried to the verge of the precipices washed by the Hongrin in the Gruyères, sawing the cabaret of Allières literally in two, and carrying away the upper story, to the amazement of the inmates, who were thus ejected from the attics to the ground-floor without a moment's notice to quit. When any accident fatal to life occurs on the Jaman, it is forbidden to remove the body until the arrival of a magistrate, excepting the mother be present, in which case her sanction is deemed sufficient. The presence of the father is not considered equal authority. There is something very touching in this deference to maternal feeling.—*Mrs. Strutt's Domestic Residence in Switzerland.*

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[Queenstown.—From a Drawing by Mrs. Simcoe, taken during the Revolutionary War.]

THE NIAGARA DISTRICT, WESTERN CANADA.—No. I

Queenstown is situated on the Niagara River, or more properly Strait, about seven miles above the Falls, and six from the shores of Lake Ontario. There is a good and pleasant road parallel to the river from Fort Erie on the lake of the same name, through Queenstown, to Newark on Lake Ontario. The length of the Niagara Strait is thirty-five miles: by this outlet the waters of Lake Erie flow into Ontario, passing in their course over the tremendous cataract. A succession of severe actions between the Americans and the British took place in 1812, 1813, and 1814, on the banks of the Niagara; and one of the most desperate occurred within two miles of the Falls. The circumstances attending this contest were peculiarly calculated to show the hateful effects of war, as they aroused all those bad passions which seem tenfold more bitter in a border-warfare, when the ties of neighbourhood and kindred are disregarded, and their obligations violated. The militia on both sides being called out, neighbours were fighting against each other —a husband against the father of his wife, and against her brothers. Every town on the frontier was destroyed, either by one or other of the belligerent parties. In October, 1812, the American and British forces encountered each other at Queenstown, which was the scene of a sanguinary contest. The spot where the English general, Sir Isaac Brock, fell on this occasion is marked by a monument erected to his memory. It is one hundred and twenty-six feet high, and stands two hundred and seventy feet above the level of the Niagara stream, which runs just below it, so that it commands a noble view, thus described by Miss Martineau, in her 'Retrospect of Western Travel:'—"To the left a prodigious sweep of forest terminates in blue Canadian hills. On the right is the American shore. There stands the village of Lewiston (opposite Queenstown), with its winding descent to the ferry. At our feet lay Queenstown, its sordidness being lost in distance, and its long street presenting the appearance of an English village. The green river rushes between its lofty wooded banks, which suddenly widen at Queenstown, causing the waters to spread and relax their speed, while making their way with three or four bends to the lake. We saw the white church of Niagara, rising above the woods some miles off; and beyond, the vast lake, its waters grey on the horizon. There was life in this magnificent scene. The ferry-boat was buffeted by the waves; groups were in waiting on either side the ferry; and teams were in the fields." The portress was an active little Irishwoman, delighted to meet any one from the "old country;" and yet some short time before some travellers (English) had thrown down a telescope belonging to her from the top of the monument, and when she asked for payment received only abuse!

About half-way between the Falls and Queenstown there is a remarkable whirlpool, of which little notice is taken in the note-books of travellers, whose attention is too much occupied by the grandeur of the Falls. The whirlpool is most probably caused by extensive cavernous hollows in the rocky bed of the river in which the waters are partially engulfed. Millions of tons of water are precipitated over the Falls every hour, and yet here the Niagara is pent up within a narrow channel not exceeding one hundred yards in width. Mr. Buckingham mentions, in his recent work on the United States of America, that "so completely is the current carried round in the circular whirlings that water assumes in any vortex having a large outlet at its base, that trees, beams, and branches of wood are carried round and round for hours in succession in its centre, sometimes descending out of sight, and re-appearing again near the same place broken into fragments. It is compared by those who have seen both to the celebrated Maelstrom of Norway, but is on a smaller scale." In Cotton's 'Tour of the Lakes' there is a harrowing account of a boat having by accident come within range of the whirlpool, and an unfortunate person being hurried round the vortex many times before the final catastrophe, while his friends on shore could render him no assistance. In the 'Penny Magazine,' No. 147, an account will be found of a

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similar accident which ended more fortunately. The Strait is so narrow at this point, that a stone has been thrown across from the American to the Canadian side, and a suspension-bridge has been projected as a means of communication between them. The rocky cliff on either side is about two hundred and fifty feet high, and the width less than that over which the bridges at Menai and Clifton are suspended. As it would overhang the whirlpool, it is thought that the cost would be reimbursed by the payments of persons visiting the spot. There is a railway from Lewiston to Buffalo.

Immediately after passing the elevated plateau of Queenstown Heights, the land shelves abruptly towards the shores of Lake Ontario, distant five or six miles, in a manner which at once arrests the attention of the geologist. The table-land, three hundred feet high, is broken by a precipice parallel to the lake. There is little doubt that this was once the boundary of its southern shore. Colonel Whittlesey, a scientific geologist and surveyor, who was officially appointed to examine this region, gives the following grounds for this supposition, which also account for the existence of the Falls. The table-land, it is to be observed, on both sides of the Niagara Strait, namely, at Queenstown and Lewiston, is level with Lake Erie. The line where it is abruptly broken is traceable for more than a hundred miles parallel with Lake Ontario, east of the Niagara, and Colonel Whittlesey thinks still farther, to the head of the St. Lawrence, at the Thousand Isles, or even to the Heights of Abraham at Quebec, and the Falls of Montmorency. "At this latter spot, and so on up the Thousand Isles above, some mighty rupture of the rocky beds beneath seems to have occurred by some convulsion of nature, and thus furnished a passage or drain for the Upper Lakes into the Atlantic. The time when this convulsion occurred must have been simultaneous with the production of the Falls of Niagara, which until then were a part of the shores of the two lakes, which here silently commingled their waters, until the sudden rupture and draining below threw the momentum of the mighty flood from the *now* table-land, and *then* lake-bed, at Queenstown, down the high precipice or naked shore, and thus excavated for itself the deep channel of Niagara river from this point to the diminished basin of Ontario. From Queenstown, the Falls, in course of time, by gradually, as they now hourly do, breaking off the shelving calcareous rock, worked their way naturally up to their present position, seven miles above, and will ultimately penetrate into Lake Erie; when another draining will take place, of Erie, Huron, and Michigan, both which latter are also doubtless diminished basins, up to the Sault St. Mary, or Low Falls, which divides these Lower Lakes from the great inland sea of Lake Superior. When that event occurs, another Niagara will in the same way be formed at this passage into Lake Superior. And so the mighty work will proceed, until our lakes, which none of them have great rivers of their own to supply the present constant draining of the St. Lawrence, and by evaporation, will shrink to minor pools, leaving, ultimately, their rich beds bare, to become the seats of civilization and of a vast population." Such are the speculations which a view of the neighbourhood of Queenstown suggests to the geologist and philosopher.

At the embouchure of the Niagara into Lake Ontario its breadth is about a quarter of a mile. The entrance is defended by two forts, one on the Canadian and the other on the American side. When Mrs. Jameson was in Canada, just before the last troubles, the British forces in the Canadian fort consisted of three privates and a corporal, with rusty firelocks and damaged guns. She mistook the fort for a dilapidated brewery. This lady gives a very charming picture of the beauties of Ontario: "This beautiful Lake Ontario!" she exclaims,—"my lake—for I begin to be in love with it, and look on it as mine! It changed its hues every moment, the shades of purple and green fleeting over it, now dark, now lustrous, now pale—like a dolphin dying; or, to use a more exact though less poetical comparison, dappled, and varying like the back of a mackerel, with every now and then a streak of silver light dividing the shades of green: magnificent, tumultuous clouds came rolling round the horizon; and the little graceful schooners, falling into every beautiful attitude, and catching every variety of light and shade, came curtseying into the bay: and flights of wild geese, and great black loons were skimming, diving, sporting over the bosom of the lake; and beautiful little unknown birds, in gorgeous plumage of crimson and black, were fluttering about the garden: all life and light and beauty were abroad in the resurrection of Nature!" This was written when the long Canadian winter was just over.

THE FELLING AND TRANSPORT OF THE PINE AND FIR.

THE *pine* and the *fir* are among the most useful forest-trees which the world produces. There is scarcely a dwelling or a ship to be found in any part of Europe or America into the building of which one of these varieties of wood does not enter; and the juices or resinous products are particularly valuable in the arts of life. Many nations, too, procure edible substances from various parts of these trees. All the varieties of the pine and the fir genera belong to one botanical order; but without entering into a description of the trees themselves, or their relation to botany or agriculture, we may collect many instructive details respecting the geographical distribution of the pine and fir forests—the mode of felling and bringing to market—the economical uses of the timber, branches, bark, &c.—and the mode of obtaining the resinous products.

The forests of Norway and Sweden are among the most celebrated of the pine and fir kind, the Scotch pine and the spruce fir being the principal varieties. Dr. E. D. Clarke says, "If the reader will cast his eyes on the map of Sweden, and imagine the Gulf of Bothnia to be surrounded by one continuous unbroken forest, as ancient as the world, consisting principally of pine-trees, with a few mingling specimens of the birch and juniper, he will have a general and tolerably correct notion of the real appearance of the country." A common mode of transporting these trees from the forests where they are cut down, to the banks of streams or rivers, is to place them on wheel-axles, one vehicle to each tree, and then to draw them by horses guided by women. In every case a path or road is taken which will lead by the shortest route to a river—such a river, in Sweden, is the Gotha, which enables the rafts of timber to be floated down to the port of Gottenburgh. There are in Norway two rivers thus employed: one, the Glomm, which terminates at Christiania; and the other, the Drammen, which flows into the sea twenty miles westward of Christiania.

Nearly all the pine and fir timber grown in Russia, Prussia, and Poland is floated down the rivers which flow into the Baltic, generally adjacent to the ports of Memel, Dantzig, Riga, and St. Petersburg. The name of the port is often given to the kind of timber which is floated down to it: thus we hear of 'Riga timber' and 'Memel timber.' It is said that the timber shipped at Memel comes principally from the estates of Prince Radzivil, in Polish Prussia; it is more abundant than that shipped at any other of the Baltic ports; but its quality is inferior to that of Dantzig; while this latter, again, yields the palm of superiority

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to the timber from Riga. This latter-named kind is largely used for the masts of English and French vessels; and in reference to it Mr. M'Culloch observes, "The mast-trade is very extensive. The burghers of Riga send persons, who are called 'mast-brokers,' into the provinces to mark the trees, which are purchased standing. They grow mostly in the districts which border on the Dnieper, and are sent up that river to a landing-place, whence they are transported thirty versts (about twenty-three miles) to the Dwina, where, being formed into rafts of from fifty to one hundred pieces each, they descend the stream to Riga. The tree which produces the longest masts is the Scotch pine. The pieces, which are from eighteen inches to twenty-five inches in diameter, are called 'masts;' and those under these dimensions 'spars,' or, in England, 'Norway masts,' because Norway exports no trees more than eighteen inches in diameter. Great skill is required in distinguishing those masts which are sound from those which are in the least degree internally decayed. They are usually from seventy feet to eighty feet in length."

Mr. Howison has given a very interesting account of the train of operations whereby Russian pine and fir are conveyed to St. Petersburg for shipment. As all the large timber near the capital has long since been cut down, the supply is obtained from a distance in the interior. A Russian proprietor wishing to dispose of the timber on his property, having completed a bargain with the St. Petersburg merchant, sets his peasantry to work in picking out, cutting down, and dragging the trees from the forests to the lakes and rivers. This work generally takes place during the winter months, in order that everything may be ready for floating the timber to St. Petersburg as soon as the ice on the rivers and lakes breaks up. As the ground is generally covered several feet deep with snow, and the trees judged to be sufficiently large and sound for the foreign market lie widely apart, the workmen and others employed in picking them out are compelled to wear snow shoes, to prevent them from sinking in the snow. When the trees are found, they are cut down with hatchets, and the head and branches lopped off. The trunk is then stripped of its bark, and a circular notch is cut round the narrow end of it, to facilitate the fixing of the rope by which the horses are to drag the trunk along; and a hole is made at the other end for a handspike, to steer the log over the many obstacles that lie in its way. Many of these trees are seventy feet in length, and of proportionate diameter; and they are drawn by from five to nine horses each, yoked in a straight line one before another, since the intricate narrow paths in the woods will not permit of their going in any other way. One man mounts upon the leading horse, and another upon the middle one, while others support and guide with handspikes the large and distant end of the tree, to raise it over the elevations of snow, and make it glide smoothly along. The conveyance of these large trees, the long line of horses, and the number of peasants accompanying them through the forests, present a very picturesque appearance. In many cases the trees are brought nearly a thousand miles before they are delivered to the merchant; and they generally remain under his care another winter, to be shaped and fitted for exportation in such a manner as to take up as little room as possible on shipboard; so that the Russian timber does not reach the foreign consumer till two years after it has been cut down. When the trees are delivered to the merchant, they are carefully examined to ascertain their soundness, and for this purpose a hatchet is struck several times against them, the emitted sound affording the means of estimating the soundness of the tree. Those which are defective, and which are called 'braake,' are about one-tenth of the whole These trees are conveyed by horses, in the manner described above, only so far as is necessary to bring them to the margin of some of the lakes or streams which have water communication with St. Petersburg, floating being then the mode of transit adopted.

In the parts of Germany bordering on the Rhine, the timber is conveyed by the quickest route to that river, and then floated down in immense rafts, which have often been described by tourists. The author of 'An Autumn near the Rhine' says, "A little below Andernach the village of Namedy appears on the left bank under a wooded mountain. The Rhine here forms a little bay, where the pilots are accustomed to unite together the small rafts of timber floated down the tributary rivers into the Rhine, and to construct enormous floats, which are navigated to Dordrecht (Dort), and there sold. These machines have the appearance of floating villages, each composed of twelve or fifteen little wooden huts, on a large platform of oak and deal timber: they are frequently eight or nine hundred feet long, and sixty or seventy in breadth." This raft is composed of several layers of trees, placed one on another, and tied together, the raft drawing not less than six or seven feet of water. Several smaller rafts are attached to the large one, besides a string of boats loaded with anchors and cables, and used for the purposes of sounding the river and going on shore. The rowers and workmen sometimes amount to seven or eight hundred, superintended by pilots; and over the whole is a proprietor or manager, whose habitation is superior to the others. The "domestic economy" of the raft is very complete. Poultry, pigs, and other animals are to be found on board, and several butchers are attached to the suite. A well-supplied boiler is at work night and day in the kitchen. The dinner-hour is announced by a basket stuck on a pole, at which signal the pilot gives the word of command, and the workmen run from all quarters to receive their rations. The consumption of provisions during the voyage from Andernach to Dort is enormous, sometimes amounting to forty or fifty thousand pounds of bread, eighteen or twenty thousand pounds of fresh meat, with salt meat, butter, vegetables, and a host of *et ceteras*. A very large capital is necessary to undertake the formation of one of these rafts.

When we proceed southerly towards Switzerland, we find pine and fir forests elevated so much above the level of the rivers, as to give occasion for no small exercise of ingenuity in devising the means of transport. A remarkable instance of the plan, now frequently adopted, was afforded by M. Rupp about thirty years ago, in reference to the means of transporting the wood from the forests on Mount Pilate to the Lake of Lucerne. The mechanism has been so often described, that a slight notice will suffice here. In the year 1810 the price of Baltic timber was so high, that a hope was entertained of bringing into profitable sale the timber on this Swiss mountain, hitherto untouched on account of the difficulty of conveyance. M. Rupp conceived the idea of making an inclined plane which should extend the whole distance from the top of the mountain to the Lake of Lucerne, about eight miles. This inclined plane consisted of a trough, formed of twenty-five thousand pine-trees, six feet broad, and from three to six feet deep. To preserve a regular slope, it had to be conducted over the summits of rocks, along their sides, through tunnels, and over deep gorges, where it was sustained by scaffolding. The trough was kept constantly moist, and the trees descended along it into the lake with extraordinary rapidity. The larger pine-trunks, about a hundred feet in length, descended through the whole distance of eight miles in six minutes; a rapidity which pro-

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duced an effect otherwise almost inconceivable to the view of a bystander. When the trunks arrived at the lake, they were floated down the Rhône to the sea. As the war, which occasioned the high price of Baltic timber, led to the construction of the Slide of Alpnach (as the contrivance was called), so did the return of peace restore the timber-trade to its old channel; and the Swiss project is understood not to have succeeded as a commercial speculation.

In many of the Alpine districts between Austria and Italy the difficulty of transporting timber to the level of the rivers has led to the construction of slides or troughs somewhat similar to that of Alpnach. In all such cases the slides are formed of several fir-trees placed side by side, and smoothed by being stripped of their bark: they are always made in such a direction as to maintain a pretty uniform slope. The slides are chiefly made use of in winter, at which time they are rendered more slippery by being wetted with water, which freezes immediately. A wood-cut and description of one of these Tyrolean districts have been given in No. 532.

In Scotland, one of the principal pine-forests is said to be that of Rothiemurchus, which spreads over the glens and valleys of the Grampian Hills. The timber from this forest is generally floated down the river Spey; and when, from a long season of drought, or any other cause, there is any difficulty in getting it down to the river, the workmen collect the trees into a suitable dell, and having built up a temporary dam, they wait the coming of a flood, which in a country of such varied surface is no rare occurrence. As soon as the temporary dam is full of water, they break down the boundary, and the liberated waters, bursting from their confinement, carry the trees with them impetuously down the Spey.

Every one who has heard of Canada and the United States must be aware that the pine and fir forests cover a vast area of the new continent. Among the remarkable features of Canada, the 'lumbering-parties' are not the least picturesque. These are clubs or bands of men who form a kind of 'joint-stock tree-cutting company,' and undergo no few hardships in the course of their labours. The proceedings of these lumbering-parties have been fully described by an eye-witness, in our No. 352.



1 Foot

[Etruscan Vase.

ETRUSCAN ANTIQUITIES AT THE BRITISH MUSEUM.

THERE perhaps is not any department of the British Museum devoted to antiquities which, to those who know comparatively little of ancient history, excites so much interest as the Egyptian room. The most unlearned will survey attentively the intelligible memorials of manners and customs which, though so ancient, exhibit feelings and passions essential to man's nature in every age. The ornaments worn three thousand years ago gratified the same feelings then as they do now. The insight obtained into the domestic life of so remote a period shows only different modes of subserving the same ends amongst a people who made the greatest advance in early civilization. The Museum catalogue affords full explanation of the various objects presented to the eyes of the visitor. In passing through a room nearly empty, we enter another apartment, the objects in which are not yet perfectly arranged, nor are they described in the catalogue. These are the Etruscan antiquities. They do not at once claim the

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visitor's attention so strongly as the memorials of ancient Egypt, but still we may read in them passages in the history of a people, and learn something of their domestic life, public ceremonies, and religious institutions. We know little besides what these vases tell us of a people who existed in Italy prior to the Romans, and who gave a deep and lasting impression to the religion and institutions of Rome. The language of the Etrurians is forgotten, the site of many of their cities is a matter of dispute; and had it not been for the great veneration in which they held their dead, and the arrangements which they made for their sepulture, we should have been unable to follow them to the remote times in which they existed as a flourishing people, evincing a taste, refinement, and magnificence in the arts and in their public ceremonials, which we can scarcely believe of a nation whose history is otherwise so obscure. They excelled in various arts, but they are universally known for their works in baked clay.

Several centuries before the building of Rome, which occurred about two thousand six hundred years ago, the Etrurians were settled in Italy on both sides of the Apennines. Etruria Proper is now the present Tuscany, with the addition of that part of the Papal States which lies on the banks of the Tiber. The Etruscan towns formed independent communities, governed by an aristocracy, the 'people,' who were probably a conquered race, being in a state of serfdom. The Etruscan settlements in Italy were established probably about three thousand or three thousand two hundred years since; and their existence as a nation was destroyed by the Romans after it had lasted eight or ten centuries. Their lands were given to the military colonies of Rome. The Etruscan language gradually became obliterated, except among the priests; and soon after the establishment of Christianity it became finally extinct. Niebuhr and all the best authorities agree that it is lost. Dr. Arnold, in his 'History of Rome,' in reference to the probability of a knowledge of it being restored, remarks that "the study and comparison of the several Indo-Germanic languages is making such progress, that if any fortunate discovery comes to aid it, we may hope to see the mystery of the Etruscan inscriptions at length unravelled." In the meantime we must be content with the pictorial language of the vases found in the tombs. These vases belong to three different periods of art, each exhibiting its peculiar style. The most ancient are those which resemble the Egyptian style; and it has been asserted that they were imported from Egypt, but they were most probably of native manufacture from Egyptian copies: harpies, sphynxes, griffins, &c. are figured upon them. They are party-coloured of red and black upon a pale yellow ground. The next in order are those with black figures on a red ground, in stiff and ungraceful outline, while the form of the vase itself is often very elegant. The most modern have red figures. The form of the vase is still more elegant, and often exquisitely beautiful, and the figures are graceful and spirited. They represent stories of gods and heroes, as well as incidents of domestic life. It has been said that the vases of this style were imported from Greece; but, on the other hand, the more probable history of the manufacture is that it was first brought from Egypt into Etruria, and was there carried to very high perfection, as shown in the black figured vases; and that it was afterwards further improved by Greek artists who settled in Etruria.

Mrs. Hamilton Gray, who is an enlightened enthusiast on the subject of Etruscan antiquities—of which she has formed a valuable collection at Bolsover Castle, containing several unique articles—remarks, in her 'Sepulchres of Etruria,' that "you will rarely see a black figure easy, natural, or graceful, however exquisite may be the beauty of its workmanship: and you will seldom be able to trace in a red figure that peculiar stiff and rigid quaintness which is characteristic of the most ancient Etruscan art. Those black figures which have a sketchy and flowing ease are on vases of a very inferior material and execution, and belong to the period of the decay of art, like the roughly-drawn red figures which are so common." The most modern Etruscan vases are about two thousand years old. None have been found at Herculaneum or Pompeii, though they were made in the neighbourhood in the highest perfection; and vases of terra-cotta, not painted, exist in great number in these lava-covered cities. It is inferred from this circumstance that the peculiarities of the Etruscan art had been lost before these places were covered by the eruption of Vesuvius in the first century.

Tarquinia, Veii, Vulci, Tuscania, and the other cities from whose necropoli the vases and other Etruscan remains have been collected, are in the neighbourhood of Civita Vecchia, and within a day's journey of Rome. The dealers in antiquities at Rome hire land where the burying-places were situated, and there carry on their excavations. The cemetery of an Etruscan city was as large as the city itself. Above two thousand tombs have been opened in that of Tarquinia, and it is computed to extend over sixteen square miles, and to contain not less than two million tombs; and yet it is surrounded on all sides by cemeteries of other cities of scarcely inferior extent. A common unpainted tomb consists of two vaulted chambers, small and low. On one side stands the sarcophagus, or bier, with its wreath, or arms, and around upon the walls are bronzes and terra-cotta. There are usually a number of vases on the ground near the sarcophagus. The subjects of the painted tombs are chariot-races, festivals, battles, in a spirited and lively-coloured style, "expressed," says Mrs. Hamilton Gray, "with a grouping and a spirit which is Greek, and a mannerism which is Egyptian." The lids of the coffins have, in some cases, figures of men and women in alto relievo, and in the coffins have been found a wreath of ivy, or of bay, in pure gold, or a helmet and spear; and in others something of gold or bronze, scarabæi, gems, jewellery; but rarely coins.



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THE TRINITY HOUSE.

IN our previous numbers we have given an account of the classification of the mercantile navy of Great Britain, and of the nature of the establishment known as "Lloyd's;" we now complete the subject by a description of the nature and duties of the corporation known as the Trinity House, to whose care are committed the lighthouses of Great Britain, the pilotage of the Thames, and other duties. The full title of the corporation is 'The Master, Wardens, and Assistants of the Guild, Fraternity, or Brotherhood of the Most Glorious and Undivided Trinity, and of Saint Clement, in the parish of Deptford Strond in the county of Kent'—an institution to whose members is intrusted the management of some of the most important interests of the seamen and shipping of England. Its duties and powers will best appear by a review of its history, and of the royal charters, grants, and several statutes under which the same exist. The earlier records, together with the house of the corporation, were destroyed by fire in 1714, so that the origin of the institution can only now be inferred from usage and the occasional mention of its purposes in documents of a later period. It seems however certain that the increase of shipping and the use of vessels of great burden having augmented the importance of a correct knowledge of the intricacies of the navigation of the channels leading into the river Thames and of the river itself, an association of seamen was formed for the purpose of forwarding and assisting the attainment of that object. It was material also that this knowledge should be solely possessed by British subjects; and probably this was present to the mind of Henry VII., who, when earl of Richmond, with a very inferior fleet, had crossed the English Channel from Harfleur, and effected a landing at Milford Haven, without molestation. That king bestowed great care upon the improvement of the navy, and it is presumed that with him originated the scheme, afterwards carried into effect by his son Henry VIII., of forming efficient Navy and Admiralty boards, which then first became a separate branch of public service. During the reign of Henry VIII. the arsenals at Woolwich and Deptford were founded; and we learn from Stowe that the Deptford-yard establishment was subsequently placed under the direction of the Trinity House, who likewise surveyed the navy provisions and stores. The earliest official document now extant is a charter of incorporation made by Henry VIII. in the 6th year of his reign. The first master acting under it was Sir Thomas Spert, commander of the famous ship called Henry Grace-à-Dieu, built by Henry VII. An exemplification of this charter was granted by George II. in the third year of his reign. By this charter the "shipmen and mariners are to establish a certain guild or perpetual fraternity;" and the brethren are empowered from time to time to elect one master, four wardens, and eight assistants, to govern and oversee the guild, and have the custody of the lands and possessions thereof, and have authority to admit natural-born subjects into the fraternity, and to communicate and conclude amongst themselves and with others upon the government of the guild and all articles concerning the science or art of mariners, and make laws, &c. for the increase and relief of the shipping, and punish those offending against such laws; collect penalties, arrest or distrain the persons or ships of offenders, according to the laws and customs of England or of the Court of Admiralty. The charter also grants to the corporation all liberties, franchises, and privileges which their predecessors the shipmen or mariners of England ever enjoyed.

It is supposed that prior to the incorporation by Henry VIII. there was a station belonging to the association of seamen near the entrance of the river, for the purpose of supplying pilots to vessels inwards, as well as one at Deptford or London for the supply to vessels outwards.

On arriving at the reign of Queen Elizabeth it is impossible not to be struck by the wisdom and foresight of the measures taken by that queen through the agency of the Trinity House for the purpose of forwarding the interests of the sea-service, measures the more to be regarded when brought into contrast with those of some of her successors. In the first year of her reign she recognised all the rights and immunities of the corporation (reciting in a charter confirming the same certain grants from King Edward VI. and Queen Mary); and in the 8th year of her reign an act was passed, enabling the corporation to preserve ancient sea-marks, to erect beacons, marks, and signs for the sea, and to grant licences to mariners during the intervals of their engagements to ply for hire as watermen on the river Thames. This act describes the members of the corporation as "a company of the chiefest and most expert masters and governors of ships incorporate within themselves, charged with the conduction of the queen's majesty's navy royal, and bound to foresee the good increase and maintenance of ships and of all kind of men traded and brought up by watercraft most meet for her majesty's marine service;" and after reciting the destruction of steeples, woods, and other marks on the coasts, whereby divers ships had been lost, to the great detriment and hurt of the common weal and the perishing of no small number of people, prohibits the destruction of any existing marks after notice, under a penalty of 100*l*., a very heavy fine in those days. An important question arose in the reign of James I., whether the words of the act of the 8th Elizabeth included *lighthouses*, which it would seem had not been introduced in England at the time it was passed: it appears to have been held by the two chief justices, 1 James I., that they did (4 *Inst.*, 149). Nevertheless, in 1616, Sir William Erskine and Sir John Meldrum having applied for a patent to erect lights at Winterton, the corporation of Trinity House petitioned against it, on the ground that they alone were entitled to make such erections, and the privy council decided in their favour. The king, however, was prevailed on to refer the matter to Sir Francis Bacon, then attorney-general, who reported his most able opinion in these striking words:—"That lighthouses are marks and signs within the meaning of the statute and charter. That there is an authority mixed with a trust settled in that corporation for the erection of such lighthouses and other marks and signs from time to time as the accidents and moveable nature of the sands and channels doth require, grounded upon the skill and experience which they have in marine service; and this authority and trust cannot be transferred from them by law, but as they only are answerable for the defaults, so they only are trusted with the performance, it being a matter of an high and precious nature, in respect of the salvation of ships and lives, and a kind of starlight in that element." This was read in council, and on the 26th March, 1617, an order was made reciting it, and "that their lordships found further cause to be confirmed in their first opinion that the masters of the Trinity House of Deptford Strond ought solely to have the erecting and disposing of all such sea-marks and signs, and that no other person ought to intermeddle therein; which their lordships did this day declare in council as the opinion of the board, with a saving still to his majesty's further pleasure. But withal straitly admonished the said masters of the Trinity House upon their duties, that as they were only trusted, and all others excluded, they should be careful to discharge

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that trust which the state had reposed in them, and that in all places needful they should cause to be speedily and timely set up such lights and other sea-marks and signs as may serve for the safe direction of sea-faring men upon any of his majesty's coasts whatsoever, that the lives, ships, and goods of his majesty's subjects, friends, and allies may not be in danger of perishing through their negligence or want of care." King James probably disapproved of this decision, because it went to preclude him from exercising that lucrative trade in the sale of monopolies and patents which formed so principal a grievance of his reign. Accordingly very shortly afterwards, Sir Francis Bacon having been made lord keeper, the same point of law was referred to Sir Henry Yelverton, then attorney-general, and such of the king's council as he might think proper to call to his assistance.

The result was the following report, more satisfactory perhaps to his majesty, but the cause, in aftertime, of much evil, loss, and expense to the nation, because the management of several lighthouses was in consequence granted to individuals. After stating the circumstances, the Report, which is dated 4th June, 1617, goes on:—

"We herein certify our opinion to your lordships:—

"1. That lighthouses are signs and marks within the meaning of the statute aforesaid.

"2. That there is an authority given by the statute to the Trinity House, to erect such lighthouses if they think fit, and a trust reposed in them to do it if they will.

"3. That they of the Trinity House cannot transfer this authority to any other.

"But we are of opinion that the authority given to the Trinity House by the statute 8th of Elizabeth, taketh not away the power and right which was and still is in the Crown by the common law to erect such houses. For that statute is made wholly in the affirmative, that they of the Trinity House shall and may erect such lights and marks at sea, but excludes not his majesty. And we are informed that since the statute, both in the time of his majesty and of the late queen, there have been some lighthouses erected by authority from the crown.

"And therefore, howsoever the ordinary authority and trust for the performance of this service is committed to the said corporation alone, as persons of skill and trust to that purpose, yet if they be not vigilant to perform it in all places necessary, his majesty is not restrained to provide them according to his regal power and justice, for the safety of his subjects' lives, goods, and shipping in all places needful."

In the 36th year of her reign Queen Elizabeth, but partly it would seem at the praiseworthy instance and by the aid of Lord Howard of Effingham, her high admiral, made a grant to the corporation of the lastage and ballastage of all ships in the river Thames, and of the beaconage and buoyage upon the coasts of the realm, which had previously afforded a considerable source of revenue to the lord high admiral. The grant recites that he had surrendered into the queen's hands the lastage and ballastage of all ships coming into or being in the Thames, and also the right to erect and place beacons, buoys, marks, and signs for the sea, on it or on the shores, coasts, uplands, or forelands near it, and besought her to grant all powers respecting these matters to them. And it then proceeds to grant the same and all fees relating to them in the fullest manner to the corporation for ever.

James I. soon after his accession granted a charter of confirmation dated 1604. What else he did has already been stated, and by him and his successors various patents for and leases of lighthouses to individuals were at different times granted. Charles II. also granted to the Trinity House a charter of confirmation, but in the 17th year of his reign he granted the right of lastage and ballastage to one Colonel Carlos. This was the more extraordinary, because by the recital in his charter of the grant of Elizabeth he recognized the right to be in the corporation: it was however conferred upon Colonel Carlos on the assurance that it would not injure them, and the colonel was to pay 1000 marks a year for it into the Exchequer. The corporation resisted this grant successfully, and soon after Colonel Carlos surrendered it to the king, who re-granted it to the corporation for 31 years (Elizabeth's grant having been "for ever"), with the addition of all the waste lands bordering on the Thames from Staines Bridge to the Medway. This portion of the grant was however disputed by the City of London, and eventually the king re-granted it as it had been granted by Elizabeth, except that the fees and profits were expressly appropriated to the use of poor seamen, their wives, &c., and the 1000 marks were reserved to Colonel Carlos. The grant confirms the exemption of the brethren and their servants, &c. from all service, civil and military, unless by order of Privy Council. James II., who was much interested in naval matters, granted a fresh charter, the one now in force, in the first year of his reign. It recites the former grant and charter, and declares the body to be a corporation, and that for the future it shall consist of one master, and one deputy master, four wardens and four deputy wardens, eight assistants, and eight deputy assistants, eighteen elder brethren, and a clerk. The master nominated by the charter was Pepys, then secretary to the admiralty. It determines the mode of election of those officers, their continuance in office, and the mode of removing them from it, if necessary; and declares that all seamen and mariners belonging to the guild shall be younger brethren. It directs the masters and wardens to examine such boys of Christ's Hospital as shall be willing to become seamen, and to apprentice them to commanders of ships. It also enables them to appoint and license all pilots into and out of the Thames, and prohibits under penalties all other persons from exercising that office: it also authorises the corporation to settle rates of pilotage, &c., to hold courts, &c., to punish seamen deserting, &c., and make laws as to their subject-matters not inconsistent with the laws of the kingdom. It also contains many provisions directed to the object of keeping the navigation of the channels secret from foreigners, and renders the officers of the corporation liable to attend when required at the king's bidding. Since that time several acts of parliament have been passed for the purpose of authorising the Trinity House to regulate matters connected with the pilotage, &c. of vessels.

The various provisions in matters of pilotage under the management of the corporation were repealed by the 6 Geo. IV., c. 125, entitled 'An Act for the amendment of the law respecting pilots and pilotage, and also for the better preservation of floating lights, buoys, and beacons,' which recites the extent of the jurisdiction of the Trinity House in regard to pilots to be upon the river Thames, through the North Channel, to or by Orfordness, and round the Long Sand Head, or through the Queen's Channel, the South Channel, or other channels into the Downs, and from and by Orfordness and up the North Channel, and up the rivers Thames and Medway, and the several creeks and channels belonging or running into the same; and contains a variety of minute regulations respecting the examination, licensing, and employment of pilots, the rates of pilotage, provisions for decayed pilots, the protection of buoys, &c. At the present time however, besides those under the jurisdiction of the Trinity

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House and the lord warden of the Cinque Ports, many independent pilotage establishments exist in various parts of the kingdom; but the expediency of subjecting all these to the sole, uniform management of the Trinity House has been felt for some time past, and will probably soon become the subject of parliamentary enactment. The inconvenience and disadvantage resulting from the exercise of similar authorities vested in the hands of different parties had been felt with regard to the lighthouses on the coast, several of which were vested in private hands by the crown; while some had been in times past leased out by the corporation itself, the lights in both instances being found to be conducted probably rather with a view to private interest than public utility. By an act therefore of the 6 & 7 Wm. IV., c. 79, passed "in order to the attainment of uniformity of system in the management of lighthouses, and the reduction and equalization of the tolls payable in respect thereof," provision was made for vesting all the lighthouses and lights on the coasts of England in the corporation of the Trinity House, and placing those of Scotland and Ireland under their supervision. Under this act all the interest of the crown in the lighthouses possessed by his Majesty was vested in the corporation in consideration of 300,000*l.* allowed to the Commissioners of Crown Land Revenue for the same, and the corporation were empowered to buy up the interests of the various lessees of the crown and of the corporation, as well as to purchase the other lighthouses from the proprietors of them, subject in case of dispute to the assessment of a jury. Under this act purchases have been made by the corporation of the whole of the lighthouses not before possessed by that body, the amount expended for the purpose being little short of a million of money.

The annual revenue of the corporation is very considerable, and is derived from tolls paid in respect of shipping receiving benefit from the lights, beacons, and buoys, and from the ballast supplied. The ballast is raised from such parts of the bed of the river as it is expedient to deepen, by machinery attached to vessels, and worked partly by the power of steam, and partly by manual labour. The remainder of the revenue proceeds from lands, stock, &c., held by the corporation, partly by purchase, partly from legacies, &c., and donations of individuals. The whole is employed upon the necessary expenses of the corporation in constructing and maintaining their lighthouses and lights, beacons and buoys, and the buildings and vessels belonging to the corporation; in paying the necessary officers of their several establishments, and in providing relief for decayed seamen and ballastmen, their widows, &c. Many almshouses have also at various times been erected, which are maintained from the same funds. The present house of the corporation is on Tower Hill; the Trinity House was formerly in Water Lane, where it was twice destroyed by fire. Of the Elder Brethren, eleven consist of noblemen and heads of the government departments, admirals, &c., who are styled honorary brethren; twenty are maritime commanders, selected from the several branches of the merchant service, who have retired from employment, and recently one has been chosen from the service of her Majesty's navy. The younger brethren (who are unlimited in number), are or have been commanders of merchant-ships. Neither the honorary members nor the Younger Brethren derive any pecuniary advantage from their connection with the corporation. The present master is the Duke of Wellington. Mr. Pitt filled that office for seventeen years, and King William IV. was master at the time of his accession to the throne. Formerly, according to Stowe, sea-causes were tried by the Brethren, and their opinions were certified to the common-law courts and courts of Admiralty, such cases being referred to them for that purpose. This is not, however, the practice at present; but two of the Elder Brethren now sit as assistants to the judge in the court of Admiralty in almost all cases where any question upon navigation is likely to arise. The various duties of the corporation are parcelled out among the wardens and different committees appointed for the purpose of discharging the same. One of the most important of these is the Committee of Examiners, before whom all masters of vessels in the navy, as well as pilots, undergo an examination. The deputy master and Elder Brethren are from time to time employed on voyages of inspection of their lighthouses and lights, beacons and buoys, not unfrequently in most trying weather and seasons; and they are also often engaged in making surveys, &c. on the coast, and reports on such matters of maritime character as are referred to them by the government. The sums paid to the deputy master and Elder Brethren for their services are—to the former 500*l.* per annum, and 100*l.* further as the chairman of all committees, and to each of the Elder Brethren 300*l.* per annum.

Power of Learning.—As for fortune and advancement, the beneficence of learning is not so confined to give fortune only to states and commonwealths, as it doth not likewise give fortune to particular persons. For it was well noted long ago, that Homer hath given more men their livings than either Scylla, or Cæsar, or Augustus ever did, notwithstanding their great largesses and donatives, and distributions of lands to so many legions: and no doubt it is hard to say whether arms or learning have advanced greater numbers. And in case of sovereignty we see, that if arms or descent have carried away the kingdom, yet learning hath carried the priesthood, which ever hath been in some competition with empire.—*Lord Bacon.*

Food best adapted for Man.—The food best adapted for man is that which contains a due mixture of azotised matter (fibrine, albumen, &c.), and non-azotised matter (sugar, starch, &c.). Dr. Liebig says:—"A nation of hunters, on a limited space, is utterly incapable of increasing its numbers beyond a certain point, which is soon attained. The whole of the carbon necessary for respiration must be obtained from the flesh of animals, of which only a limited nubmer can find food on the space supposed. But 15 lbs. of flesh contain not more carbon than 4 lbs. of starch; and while the savage, with one animal and an equal weight of starch, could support life and health for a certain number of days, he would be compelled, if confined to flesh alone, in order to procure the carbon necessary for respiration and for the animal heat, to consume five such animals in the same period. It is easy to see, from these considerations, how close the connection is between agriculture and the multiplication of the human species. The cultivation of our crops has ultimately no other object than the production of a maximum of those substances which are adapted for assimilation and respiration in the smallest possible space. Grain and other nutritious vegetables yield us, not only in the form of starch, &c., the carbon which protects our organs from the action of oxygen, and serves to produce also the heat essential to life, but also, in the form of vegetable fibrine, albumen, and caseine, our blood, from which all the other parts of the body are developed. Man, when confined to animal food, respires, like the carnivora, at the expense of the matters produced by the metamorphosis of organised tissues; and, just as the lion, tiger, and hyena, in the cages of a menagerie, are compelled to accelerate the waste of the organised tissues by incessant motion, in order to furnish the matters necessary for respiration and for animal heat, so the savage, for the very same object, is forced to make the most laborious exertions, and to go through a vast amount of muscular exercise. He is compelled to consume force, merely in order to supply matter for respiration. Cultivation is the economy of force. . . . The unprofitable exertion of power, the waste of force in agriculture, in other branches of industry, in science, or in social economy, is characteristic of the savage state, or of the absence of cultivation."—*Quarterly Review.*

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